A BUILDING IS NOT ITS SILHOUETTE: STRUCTURAL BLINDNESS IN 3D GENERATION EVALUATION

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Paper under double-blind review

ABSTRACT

Generative models of discrete 3D structure are evaluated by rendering a few exterior views, embedding them with a pretrained encoder, and measuring a distribution distance to real data. That protocol cannot see whether a structure has an inside. The inside is what makes a build usable rather than decorative: a Minecraft house someone walks through, a LEGO model someone assembles, a game asset that needs navigable space and a collision volume.

We measure the cost with a controlled corruption. Filling every sealed void in a corpus of real buildings adds a median of 33% more blocks and leaves the exterior untouched. A multi-view DINOv2 kernel distance moves 0.14 of the spread between two samples of real data, placing the corrupted corpus and genuinely held-out data in one significance group. A kernel distance on voxel descriptors moves beyond 200 of that same unit. Seven render-space statistics, five encoders, view budgets up to sixteen views and a second domain all miss it in both directions, while resolving every corruption that reaches the exterior. Cutaway views recover it with the same encoder and the same statistic, which places the failure in the view protocol rather than in the features. Under an opaque renderer the blindness is exact: filling sealed voids changes no exterior view, so no encoder and no statistic separates it at any sample size.

We then supply what the protocol is missing. A metric earns the right to rank models only after it resolves corruptions of known severity, under gates on ordering, invariance and sample-size stability, with separate disqualification gates for memorization and mode collapse. V-KID, a kernel distance on D4-invariant voxel descriptors, passes those gates, runs on a CPU, and closes the blind spot the renders leave.

1 INTRODUCTION

Many of us are taught at a young age that we shouldn't judge a book by its cover. A building can be examined the same way. It may have rooms, doors–an interior. And, in many cases, structures rely on these interior layouts to make them usable rather than decorative. Consider a Minecraft build someone walks through, a LEGO model someone assembles, or an indoor environment that needs navigable space and collision volumes for robotic training.

Generative models of discrete 3D structure have advanced quickly, over LEGO assemblies (Pun et al., 2025; Xu et al., 2025; Kulits & Schmid, 2026), sparse voxel categories (Jung, 2025), Minecraft worlds (Merino et al., 2026) and serialized geometry (Wang et al., 2024; Sudhakaran et al., 2023). Many of their evaluation techniques come from image generation where they render a structure from a handful of views, embed the renders with a pretrained encoder such as DINOv2 (Oquab et al., 2024), and report a Fréchet inception distance (FID) (Heusel et al., 2017) or kernel inception distance (KID) (Bińkowski et al., 2018) between the generated and real feature distributions. We call this family *render-and-embed*. Its limitation is already on the record. The largest generative study in this domain (Merino et al., 2026) lists among its own limitations that render-based FID "struggles to assess internal structure (e.g., whether a building has a coherent interior)," then ranks models with it for want of an alternative. What the field lacks is a measurement of how large the effect is and a metric that sees what it misses. We supply both.

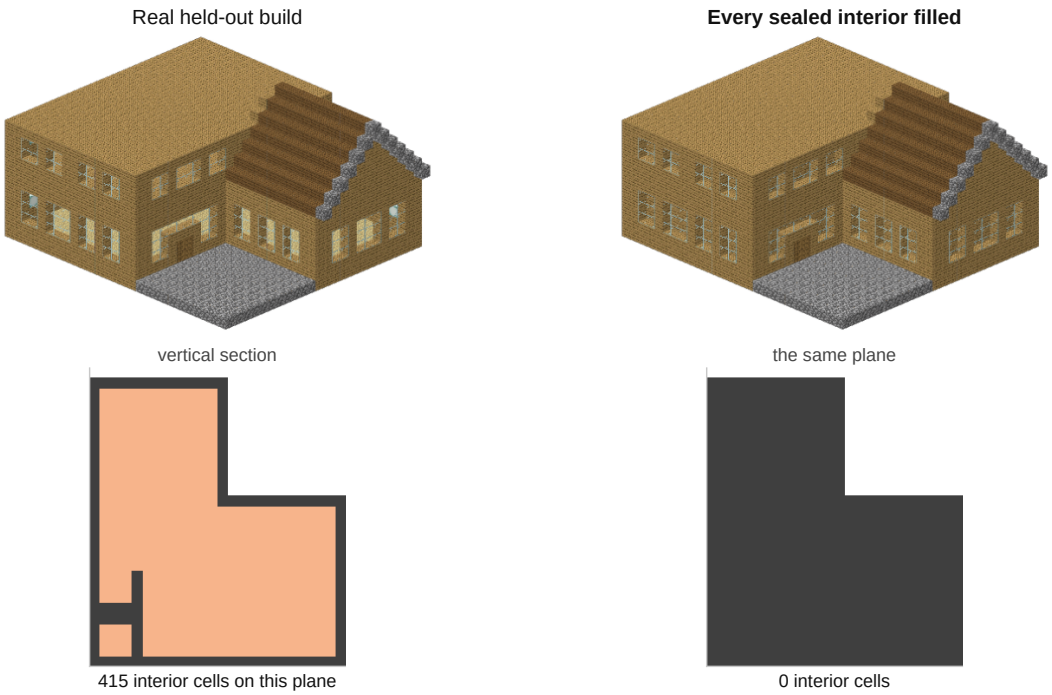


Figure 1: **A building is not its silhouette.** *Left:* a real build, sectioned vertically through its sealed interior, shown in peach. *Right:* the same build under FILL-INTERIOR, every sealed void filled by its own dominant material and cut on the same plane, almost indistinguishable in all four views the metric renders. MV-KID separates the two by 0.14 of its own noise floor, which leaves them in one significance group, while V-KID separates them by 202. Renders use the metric's own texture pack; published builds come from 3D-Craft (Appendix Q).

Our probe, FILL-INTERIOR, takes real buildings and fills every enclosed air cell with that build's own dominant material, so no new material betrays the corruption (Figure 1). It adds a median of 33% more blocks, alters 94% of examined builds, and leaves the exterior silhouette untouched. Scored at $n{=}128$ against 399 held-out real builds, the metric cannot tell the solidified corpus from genuinely held-out real data. Comparing every pair of arms and correcting for the number of comparisons (Holm, 1979), the two are statistically indistinguishable. The metric is not simply insensitive: on the same builds it separates a permutation of the block materials, which moves no geometry at all, by $+231.52$ noise floors (Table 2).

The failure is in what the views show, not in what the encoder computes. Change only the cameras, so that the renders expose the interior, and the same probe separates at 7–56 noise floors (§6.2). Under an opaque renderer it stops being a matter of degree: filling sealed voids cannot change any exterior view, so no encoder and no statistic can separate the two at any sample size (§6.3, Appendix K).

No metric should rank models until it has shown that it detects damage it ought to detect. We therefore score candidate metrics against corruptions of known severity and require monotonicity, invariance and sample-size stability before any of them orders a scoreboard, with separate disqualification gates for memorization and mode collapse. Measuring geometry directly, alongside the renders rather than instead of them, closes the blind spot. We make three main contributions.

1. **Interior blindness, and it belongs to the representation** (§6.1–6.3). Render-and-embed distances cannot distinguish a corpus of real buildings from the same corpus with every interior filled, while the same test resolves every corruption that reaches the exterior. Seven render-space statistics, five encoders, view budgets up to sixteen views and a second domain all fail the same probe, and cutaway renders recover it with the same encoder and the same statistic.

2. **A geometric metric with the complementary blindness** (§4). V-KID is a kernel distance on D4-invariant voxel descriptors. It separates the probe by 202 noise floors, needs no GPU, and is *exactly* invariant to material permutation, so neither metric alone ranks both corruptions correctly.
3. **A validation protocol, and a suite that implements it** (§4). Gates on ordering, invariance and n-stability decide whether a metric may rank anything at all. Fréchet distance, the functional form FID uses, fails n-stability, as do coverage, recall and the biased maximum mean discrepancy (MMD) estimator (Jayasumana et al., 2024); precision and density (Naeem et al., 2020; Kynkäänniemi et al., 2019) fail monotonicity.

2 Related Work

Render-based evaluation of discrete 3D generation. Recent systems for generating discrete 3D structure evaluate their outputs through rendered views, whether they emit brick assemblies (Pun et al., 2025; Xu et al., 2025; Kulits & Schmid, 2026; Ni et al., 2026), voxels (Chen et al., 2019; Jung, 2025; Awiszus et al., 2021; Earle et al., 2024) or whole Minecraft worlds (Merino et al., 2026). Dream-Cubed (Merino et al., 2026) is the largest of these. It scores generated worlds with a render-based FID, names the limitation we measure in its own limitations section, and reports raters agreeing with that FID only 54.3% of the time. FID (Heusel et al., 2017) and KID (Bińkowski et al., 2018) are the default two-sample statistics. At our sample sizes the unbiased MMD estimator (Gretton et al., 2012) is preferable, and §4 shows that a widely copied biased variant ranks by sample size instead. Precision and recall (Kynkäänniemi et al., 2019), density and coverage (Naeem et al., 2020) and CMMD (Jayasumana et al., 2024) vary the statistic and the features but stay in image space, so all inherit the blindness we measure.

Alternatives to image-space evaluation. A natural alternative to rendering is to evaluate generated geometry directly. Point-cloud methods compare shape distributions in geometric space (Achlioptas et al., 2018), avoiding some of the information loss introduced by projection to images. However, changing the representation does not by itself guarantee sensitivity to the property of interest. Luan et al. (2026) similarly identify limitations of render-based evaluation and propose surface-point sampling to improve view coverage. Their representation remains restricted to the observed boundary: in our setting, a structure and its interior-filled counterpart have identical exterior surfaces, and therefore receive identical surface-point representations. Thus, this approach addresses incomplete *view coverage* but not blindness to geometry that does not alter the exterior surface. Our geometric evaluation instead operates on volumetric structure, allowing it to detect the interior changes introduced by our controlled corruption.

Auditing generative-model metrics. Several works have argued that generative-model metrics should be tested for desirable properties rather than treated as universally valid measures of quality. Borji (2019) identifies sensitivity to meaningful distortions and invariance to irrelevant transformations as desirable metric properties, while Chong & Forsyth (2020) analyze the dependence of FID on sample size. O'Bray et al. (2022) provide a closer methdological precendent by applying controlled perturbations of increasing severity and testing whether a metric responds monotonically. They use these, however, to select metric hyperparameters where we, instead, use controlled structural corruptions as a precondition for evaluation. We examine how a metric must demonstrate sensitivity to relevant structural changes and invariance to irrelevant ones before being used to compare generators. Hervé & Salge (2021) provide a closer in-domain example of metric evaluation for procedural content generation, while Caytuiro & Sipiran (2025) audit whether generated structures exhibit a particular structural property. Our focus differs in that we audit the *observation and representation pipeline itself*: specifically, whether an evaluation protocol can detect changes to 3D geometry that are invisible in its input views

Memorization and human evaluation. Other work addresses complementary problems that arise when evaluating generative systems. Memorization can produce deceptively strong realism scores (Carlini et al., 2023; Bai et al., 2021), motivating nearest-neighbor retrieval and held-out comparisons for 3D generation (Pu et al., 2025; Meehan et al., 2020). Human judgments provide another form of validation; GDMC (Salge et al., 2018), for example, evaluates individual generated settlements against human-defined criteria. We use memorization checks and human judgments as

complementary safeguards rather than as alternatives to our structural audit. In particular, human evaluation allows us to test whether the structural information missed by automated metrics is also unavailable to observers given the same exterior views.

3 Background and Problem Setting

A *structure* here is a map $x : \{1, \dots, d\}^3 \to \mathcal{V}$ from a voxel grid to a categorical block vocabulary $\mathcal{V}$, with one symbol reserved for air. An *arm* is a finite set of structures with a name and a provenance record: a generative model's samples, a procedural generator's output, or real data under a known corruption. Evaluation compares an arm's empirical distribution Q against a fixed reference P of real structures. Two features of this setting differ from image generation. Sample sizes are in the hundreds rather than the tens of thousands, and arms may differ in size, so a statistic whose null depends on n ranks by sample size.

Both metrics we report are kernel two-sample distances. For a feature map ϕ and a kernel k,

$$\mathrm{MMD}^2(P, Q) = \mathbb{E}\, k(\phi(x), \phi(x')) + \mathbb{E}\, k(\phi(y), \phi(y')) - 2\,\mathbb{E}\, k(\phi(x), \phi(y)), \tag{1}$$

with $x, x' \sim P$ and $y, y' \sim Q$ (Gretton et al., 2012). The unbiased estimator of Eq. 1 omits the diagonal terms $k(\phi(x_i), \phi(x_i))$ from the within-set means. The biased estimator keeps them, so its null on two disjoint real halves decays with n. KID (Bińkowski et al., 2018) is the unbiased estimator with a polynomial kernel on encoder features, and FID (Heusel et al., 2017) fits a Gaussian to each feature set instead. A *render-and-embed* metric is any of these with $\phi = (\text{encoder}) \circ (\text{render})$, so the structure reaches the statistic only as a set of images.

Raw distances across features can differ by orders of magnitude, so we report all distances in *noise floors*. For each feature and sample size n, the noise floor is defined as the standard deviation of the distance obtained from repeated real-vs-reference draws under the same experimental run. Thus, a value of 1.0 indicates that the observed distance is one standard deviation of the metric's sampling variability. Because the sampling variability changes with n, the noise-floor unit is n-dependent (Appendix N). Comment: check all appendix refs

We treat a metric as *valid* for ranking when it has three properties.

1. **Ordering**: its value rises monotonically with the severity of a known corruption.
2. **Invariance**: its value does not move under a transformation that leaves quality unchanged.
3. **n-stability**: its null on real-vs-real draws does not drift with n.

Sensitivity, how *much* damage a metric resolves at a given n, is statistical power rather than measuring the right quantity, so we record it alongside the three rather than as a fourth.

4 Metric Validation

4.1 Probes

Fill-Interior fills every enclosed air cell of a real build with that build's own dominant material, which keeps the silhouette bit-identical and adds only interior mass. Under a translucent renderer it does change pixel values inside that silhouette (§6.3) and can only be observed externally when a structure contains this transparency (e.g. a window). On 200 example real houses, it adds a median of $\mathbf{33}\%$ blocks (mean 37%) and alters $\mathbf{94}\%$ of these builds. The build's own material keeps the corruption from showing as a palette anomaly. The probe reaches only *sealed* air, air unreachable from the bounding box without passing through solid, so air reachable through a doorway survives; per build, the sealed share of a build's own interior volume has a median of 100% and a mean of $\mathbf{85.5}\%$ (Appendix E).

Hollow-Shell is the converse. It keeps a block only if one of its six faces borders air 6-connected to the outside, which makes it the exact set-complement of what Fill-Interior fills, and it removes a median 13.4% of blocks from 98.4% of builds. Four further controls damage appearance or diversity rather than geometry. Material-Shuffle permutes the block multiset across a build's own occupied cells, preserving geometry exactly, and Monochrome repaints every block with one

material. TRAIN-COPY (verbatim training builds) and SINGLE-MODE (one build repeated n times) damage nothing a distribution distance objects to.

4.2 MV- AND K-KID

Both realism metrics are the unbiased estimator of Eq. 1 and differ only in ϕ (Figure 8). **MV-KID** is our stand-in for the render-and-embed family with four orthographic views of each structure at 224px, embedded with DINOv2-B (Oquab et al., 2024). **V-KID** applies the same estimator to a 77-dimensional descriptor read directly off the voxel grid, examining the distribution over the 55 D4 orbits of $2 \times 2 \times 2$ occupancy patterns, a 5-bin wall-thickness histogram, an 8-bin normalized height profile, and 9 global scalars, at about a minute per arm at n=64 on CPU (Appendix G.2). V-KID reads occupancy only, so under MATERIAL-SHUFFLE and MONOCHROME its value is *bit-identical* to the real row. It measures what the render tier cannot, and not structural quality outright.

4.3 BLOCKING GATES

Every candidate metric is scored on rungs of known damage, in units of its own noise floor. A metric that fails ordering, invariance or n-stability is reported as *gated*, with the failing gate named, and may not rank anything. Applying this to the seven statistics of Table 11 and the biased MMD estimator settles three things (Appendix G). First, we clearly see the render metric reads appearance rather than structure as MATERIAL-SHUFFLE preserves geometry exactly and is yet among the worst-scoring rungs, beaten only by MONOCHROME. Second, the biased MMD estimator is unusable at variable n: its null on two disjoint real halves reads 0.289 at n=16, 0.071 at n=64 and 0.014 at n=400, while the unbiased estimator reads 0.000 at every n. A leaderboard mixing arm sizes therefore ranks by sample size. Third, precision and density fail monotonicity under block noise, and Fréchet distance, recall and coverage all fail n-stability. Only the unbiased KID and MMD-RBF clear each condition.

4.4 AGGREGATION AND DISQUALIFICATION

The gates decide which metrics may speak. Turning several that pass into one ranked number is a separate and more arbitrary problem. BlockScore is *one* instantiation: the **maximum** over five pillars, each divided by its own in-run scale, so there are no weights to tune. The property we defend is the maximum rather than the mean. The two tiers are blind in complementary directions (§6.4), so averaging lets an arm buy a catastrophic pillar with a good one, and the trade the render tier invites is the dangerous one: perfect surfaces over a solid interior. Appendix M records two properties of the implementation: three of the five pillars are in true noise floors and two are in a coarser unit that never determines the maximum, and the null of a maximum is not zero.

Novelty and diversity are **disqualification gates** rather than weighted terms. An arm that trips one is refused a rank and shown with its reason, so a model cannot offset memorization with realism. The risk is concrete here: a pre-split autoregressive checkpoint from prior work duplicated roughly 31% of training builds at intersection over union (IoU) ≥ 0.95. Every arm must also declare the split it trained on, the requirement §6.5 turns on. Gates need auditing as much as metrics do (Appendix F).

5 EXPERIMENTAL SETUP

Our testbed is Minecraft houses: discrete 3D structures over a categorical block vocabulary, at a scale where a real corpus exists and interiors are semantically meaningful. We pool three public sources onto a shared vocabulary: a GrabCraft scrape, the 3D-Craft house traces (Chen et al., 2019; Gray et al., 2019) and the text2mc dump. A quality gate and variant-aware deduplication leave **2,661** houses in a 32^3 box (Appendix A). How that corpus is divided matters more than how it was assembled.

Public build corpora ship sibling structures, such as "American Middle Class House 10 / 22 / 9", and a random row split scatters a family across train and validation. Over 20 seeds, a mean of **28.7%** of validation items share a build family with training (s.d. 2.7, range 23.1–35.8). Leakage at that rate inflates the held-out floor and deflates every memorization statistic at once. Our canonical split is group-aware over 2,054 families at 0% cross-split family overlap, pinned and released: train 1,863

/ validation 399 / test 399. The calibration draw needs the same care, since taking the first n test builds rather than a shuffled sample inflates the measured noise floor by $\mathbf{10.4}\times$ (Appendix C).

The suite scores three kinds of arm against one fixed real reference (Table 1). *Probes* are real data with a known corruption (§4.1), *baselines* are five seeded generators and *submissions* are generative models. Baselines are fitted on the training split only, since one that had seen validation or test would hold an unearned advantage.

Table 1: The arms, used by these names throughout; recipes in Table 8.

probes (real data, corrupted)		**baselines** (procedural)	
REAL	the floor: held-out real	RANDOM-VOXELS	random voxels at real density
FILL-INTERIOR	every interior cell filled	HOLLOW-BOX	hollow box, walls and roof
HOLLOW-SHELL	interior blocks removed	GABLED-HOUSE	box, pitched roof, door, windows
MATERIAL-SHUFFLE	blocks permuted, shape kept	PATCH-STITCH	8^3 patches, different builds
MONOCHROME	one material, shape kept	TRAIN-COPY+NOISE	a training build, 10% displaced
TRAIN-COPY	verbatim training copies	**submissions** VOXEL-AR, VOXEL-AR-U, LLM-FT, LLM-FT-GEO	
SINGLE-MODE	one build repeated n times	(two token models, two language models)	

The four submissions are this project's own models. VOXEL-AR and VOXEL-AR-U are autoregressive transformers over merged-piece tokens, differing in whether builds are canonically oriented. LLM-FT and LLM-FT-GEO are LoRA fine-tunes of a 1.5B code model (Qwen2.5-Coder), the second with a relative-3D attention bias. Both decode under a 1200-token cap, and **that cap limits them**: several completions are truncated into structures one block thick (Figure 15, Appendix C). They are weak baselines.

Every control is re-scored in every run rather than cached, because a distance means nothing until a floor has been measured under *this* run's corpus, split, n and reference size. Unless stated otherwise, arms are scored at n=128 against the 399-build test reference, and significance comes from a paired bootstrap with Holm correction (Holm, 1979) over all arm pairs. The encoder ablation uses n=256 against a 512-build reference, with the noise floor taken from 48 fresh real draws. The second domain is 47,389 LEGO brick abstractions of ShapeNetCore objects on a 20^3 grid, 44% of which contain a sealed void, scored at n=256 against a 450-build reference. Its license is unresolved, so we measure on the corpus and publish no pictures of it (Appendix Q).

6 RESULTS AND DISCUSSION

6.1 THE RENDER METRIC IS FLAT ON THE INTERIOR AXIS

A metric that is merely insensitive to interiors still moves when the interior changes. MV-KID does not. On the occupancy-noise and mixture axes it is monotone over 121–143 noise floors. On the interior axis its range end to end stays under half a noise floor, 0.48 at n=64 and 0.34 at n=128, where V-KID reads 133 and 287 on the identical rungs (Appendix G). Raising n multiplies the geometric signal by 2.2 and does not lift the render signal off its sampling floor. Both metrics rise on the mixture axis, a real arm progressively replaced by bad samples, which is how a generator actually fails.

Figure 2 places both metrics on eight rungs of known damage in one run. The flatness is not a property of the one statistic we report. Table 11 puts seven render-space statistics on the same rungs. Six of them place the deletion of a fifth of a structure at 7–14 noise floors and one-minus-recall places it at 1.4, while none of them places filling the interior above 2.1. The blindness belongs to the representation they read rather than to any one of them.

6.2 IT IS THE VIEW PROTOCOL, NOT THE ENCODER

The finding could indict DINOv2 at four views rather than render-and-embed as a class. It does not (Figure 3, Table 12). Four other encoders land in the same 0.04–0.18 band as our DINOv2-B: CLIP ViT-B/16 and B/32, ImageNet-supervised ResNet-50, and the Inception network FID itself uses. Quadrupling the cameras does not help either. **Sixteen views covering roof and underside separate the probe *less* well than four** ($+0.08$ against $+0.18$), which is what added views carrying

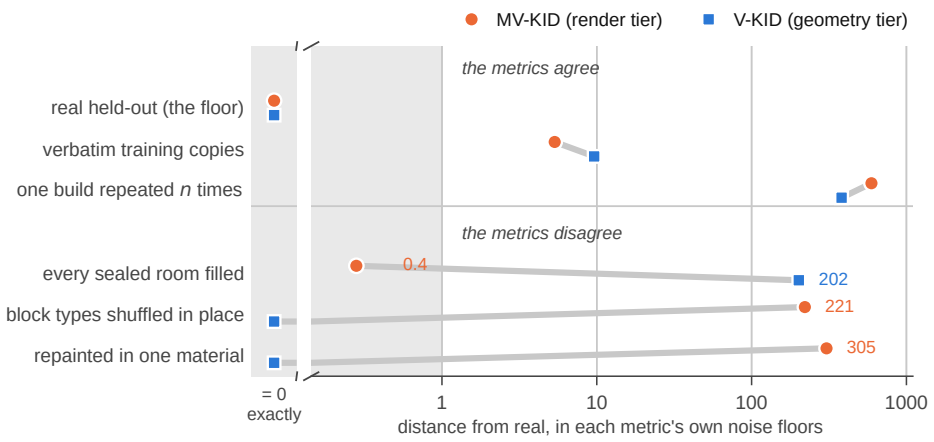


Figure 2: **Two metrics, one run, eight rungs of known damage.** Each rung's distance from held-out real, in that metric's own noise floors (n=128, symlog). The geometric metric's two material rungs are bit-identical to real, not merely small, so they sit in a separate gutter. The band marks one noise floor; per-statistic numbers are in Table 11.

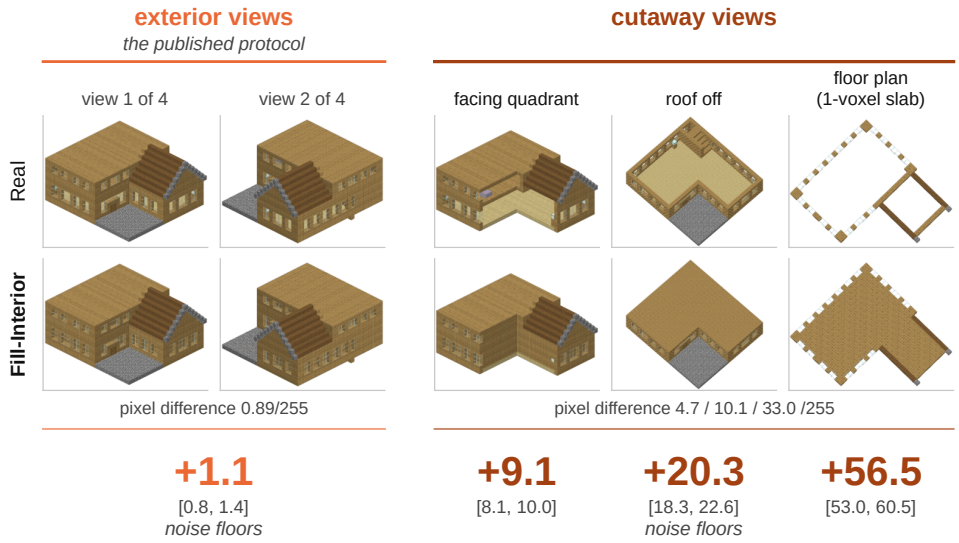


Figure 3: **The same encoder and statistic, only the views changed.** Each pair is a held-out build over its FILL-INTERIOR twin. *Left:* two of the four exterior views the standard protocol renders, indistinguishable for this build at a mean pixel difference of $0.89/255$. *Right:* three cutaway protocols on the same builds, each with its separation in noise floors. DINOv2-B, unbiased KID, n=256 against 512 reference builds; full ablation in Table 12.

no information about the quantity should do. Perspective projection reads $+0.06$. Nothing measured on exterior views reaches a quarter of one noise floor.

The cutaway rows are the positive control. The same encoder and the same statistic recover the interior at up to fifty times the separation once the view protocol stops hiding it, which makes this a statement about *information* rather than about features: **the standard view protocol does not put interior structure into the image, so no encoder downstream of it can score that structure.** A cutaway is not itself a good metric, because a section view discards the facade the render tier exists to read.

Table 2: **The leaderboard.** Appearance and geometry are in noise floors of that pillar's own in-run real-vs-real spread, and BlockScore is the **maximum** across five pillars. The three pillars that never determine the maximum here use a coarser in-run scale and are reported in Appendix G. † marks the contaminated checkpoint of §6.5, unrankable and shown for comparison.

	arm	BlockScore ↓	worst pillar	appearance	geometry
	REAL *(anchor)*	0.00	—	+0.00	+0.00
submissions					
†	VOXEL-AR *(contaminated)*	106.87	geometry	+69.76	+106.87
1	VOXEL-AR *(clean)*	**136.73**	geometry	+85.31	+136.73
2	LLM-FT	149.19	appearance	+149.19	+137.72
3	LLM-FT-GEO	191.88	appearance	+191.88	+145.58
4	PATCH-STITCH	246.15	appearance	+246.15	+183.03
5	GABLED-HOUSE	300.15	geometry	+287.97	+300.15
6	HOLLOW-BOX	476.22	appearance	+476.22	+312.39
7	RANDOM-VOXELS	603.09	appearance	+603.09	+344.94
probes and controls: known damage, not submissions					
	FILL-INTERIOR	201.80	geometry	**+0.14**	**+201.80**
	MATERIAL-SHUFFLE	231.52	appearance	+231.52	**+0.00**
	MONOCHROME	297.95	appearance	+297.95	+0.00
	TRAIN-COPY+NOISE	302.73	geometry	+153.53	+302.73
disqualified: refused a rank, with the reason					
	SINGLE-MODE	DQ	mode collapse, +45.1 spreads on voxel diversity		
	TRAIN-COPY	DQ	memorization, +10.7 spreads on retrieval percentile		
	VOXEL-AR-U *(clean)*	DQ	memorization, +3.0 spreads, at duplicate rate exactly 0.000		

6.3 BOTH DIRECTIONS, A SECOND DOMAIN, AND A THEOREM

The corruption this field actually produces is the converse of FILL-INTERIOR: the hollow shells that surface-fitted mesh and implicit generators emit. It is equally invisible. In the converse-probe run at n=128, MV-KID places HOLLOW-SHELL at +0.4 noise floors and FILL-INTERIOR at +1.0 against a real-vs-real floor of +0.0, while V-KID places the same two arms at +156.0 and +286.7. The render metric is blind along the whole interior axis rather than to one hand-built probe. Under our translucent renderer neither direction separates at the protocol's n=128. Each run estimates its own noise floor, which is why FILL-INTERIOR reads 286.7 here and 201.80 in the leaderboard run: separations compare within a run (Appendix N).

The pattern is not a property of Minecraft houses. On the LEGO corpus at n=256, with the domain pictured in Figure 6, MV-KID places the solidified arm at $\mathbf{-0.01}$ noise floors, *below* the real floor, while V-KID places it at $\mathbf{+118.6}$. On the same builds at the same n the render metric separates MATERIAL-SHUFFLE at $\mathbf{+140.0}$ floors and MONOCHROME at $\mathbf{+685.8}$, so it resolves appearance damage on this corpus and still cannot see the interior.

Whether interior damage reaches the image at all depends on the renderer. Under an *opaque* renderer it provably does not: filling sealed air cannot alter any exterior view, so no encoder and no statistic can separate it at any sample size. Rendered opaquely, all 1,024 held-out (build, view) pairs we test are bit-identical, and Appendix K gives the proof.

Our default renderer is not opaque, and under it FILL-INTERIOR alters $\mathbf{91}\%$ of the 1,596 test-split views, touching a median 1.2% of pixels at a median maximum channel difference of $103/255$. What leaks is *translucency*, not doorways: air reachable through a doorway is never sealed, whereas a glass block in front of a filled void does show the fill. Every changed pixel lies inside the silhouette, and the silhouette is bit-identical (Appendix K).

6.4 COMPLEMENTARY BLINDNESS FORCES THE AGGREGATION RULE

Table 2 shows the complementarity directly. FILL-INTERIOR reads +0.14 on appearance against +201.80 on geometry, a factor of 1,441 between two pillars scoring the same arm, and MATERIAL-SHUFFLE is the converse, +231.52 on appearance against *exactly* +0.00 on geometry.

Each tier is fooled by the corruption aimed at it and rescued by the other, which is why BlockScore takes the worst pillar. The two failures differ in severity: the best attack we have on V-KID is PATCH-STITCH, which buys near-perfect local occupancy statistics with globally incoherent structure and improves its geometry pillar over its appearance pillar by only $1.3\times$ ($+183.03$ against $+246.15$).

V-KID does not separate FILL-INTERIOR through the added mass, nor through the sealed-interior scalar the probe drives to zero. Deleting the explicit size term moves the separation from 201.8 to 202.2 floors, deleting the sealed-interior scalar moves it to 197.1, and deleting both leaves 197.3. No single block of the descriptor reproduces the result alone, the best being the scalar panel at 164.5 (Appendix B).

The sharper test is a control matched on the quantity in question. GROW-SHELL dilates the solid outward until it has added *exactly* the number of blocks FILL-INTERIOR adds to that build, which it achieves to within one block on 128 of 128 builds at a median of 196. Same builds, same material, same mass, and the only difference is whether the new mass goes inside or outside. The two metrics then part company completely. In that run MV-KID places GROW-SHELL at $+\mathbf{102.3}$ noise floors and FILL-INTERIOR at $+\mathbf{0.3}$, while V-KID places them at 214.5 and 201.8. A metric that counted blocks could not produce that pair of rows, because the block count is the one thing the two arms share.

The two tiers disagree about real, uncorrupted builds as well, with no probe involved: ranking the same 128 held-out builds by each metric's exact per-build contribution gives two nearly independent orderings, and only the geometric one tracks how much interior a build has (Figure 7). Separately, because these statistics are squared, an arm that fails on a quarter of its builds scores only 6.4% of the way to fully bad; reporting the square root recovers the failure rate to a mean absolute error of 0.013 against 0.128 (Appendix L).

6.5 CASE STUDY: CONTAMINATION NO STATISTIC SEES

The results above use corruptions we built, where ground truth is known by construction. This case study asks whether the suite catches a failure we did not build. A generator we control is the right subject, because its training split is knowable. The suite requires every arm to declare that split. VOXEL-AR, the top-ranked submission, declared one that contained 346 of the 399 canonical test builds (87%). LLM-FT and LLM-FT-GEO declared an earlier $90/10$ shuffle holding 53 of the 399 (13%). No distribution statistic sees either. VOXEL-AR's duplicate rate is 0.000 and its retrieval percentile 0.496 against a real held-out floor of 0.540, far from the verbatim-copy control at $+10.7$ spreads.

Retraining VOXEL-AR on the canonical split, with architecture and hyperparameters unchanged, isolates the effect (Figure 5). We report raw V-KID here, because the contaminated and clean runs do not share a noise floor. Contaminated, the arm scores 11.62 and separates from one language-model baseline at Holm 5.5×10^{-3}; clean, it scores 14.898 against their 15.006 and 15.869 and separates from neither. Under a correctly calibrated test it never held a significance group alone (Appendix O). Contamination lowers an arm's distance from real, so the baselines' own contamination flatters them and we report that comparison as not separated. The gate also caught the sibling: VOXEL-AR-U, retrained on the same clean split, is disqualified at $+3.0$ spreads on a retrieval percentile of 0.369 against a real floor of 0.540, at a duplicate rate of exactly 0.000.

The result bears on the paper's argument twice. What detected both contaminations was a required field rather than a detector: every memorization gate passed and the declaration did not, so benchmarks in this domain should require that declaration and refuse to rank arms that cannot make it. The second bearing is on aggregation. The clean VOXEL-AR is geometry-limited, $+136.73$ geometry against $+79.90$ appearance, and both language-model arms are appearance-limited, so a mean over pillars would have reported them as one model at two quality levels (Appendix G).

6.6 HUMAN EVALUATION

Synthetic corruption shows only that the metrics order damage we built ourselves. Human judgment is the other check. Twenty raters, naive to the hypothesis, each compared 48 label-stripped pairs rendered exactly as the metric renders them, yielding 912 judgements after one session was excluded

on catch trials. Three predictions were recorded in the instrument before any rater saw a page (Appendix H).

The corruption is invisible to people too. On exterior views raters preferred the real build over its solidified copy on 43 of 84 trials, 0.512 $[0.407, 0.616]$: chance. This decides what kind of failure this is. Had people separated what the metric could not, the metric would be failing to read something a reader reads. They cannot, so the information is absent from the picture rather than discarded by the statistic. **The same people separate the same builds once the view exposes the interior**, 69 of 90, 0.767 $[0.669, 0.842]$, which turns "nobody can tell" into "nobody can tell *from this view*". The converse probe is also at chance, 0.606 $[0.437, 0.753]$ on 33 trials.

The pre-registered pair reversed, unanimously. Both metrics rank PATCH-STITCH above GABLED-HOUSE; raters chose GABLED-HOUSE on 93 **of** 93 trials $[0.960, 1.000]$. The geometric scalars say why, and neither side is making an error. GABLED-HOUSE is a genuine statistical outlier, a hollow and almost perfectly symmetric shell at 0.562 open interior ratio against a real 0.129, so the metrics are right that it is far from the real distribution. PATCH-STITCH is assembled from patches of real builds, so its local statistics are real by construction and only its global structure is wrong, which its largest-connected-part ratio of 0.866 records against a real 0.978 (Table 10). A distribution distance over local features therefore rewards *statistically* house-like over *recognizably* a house.

6.7 LIMITATIONS

We measure two corpora and *one renderer*. The renderer is the sharper gap, and the invisibility is a theorem about opaque rendering rather than a fact about our rasterizer (Appendix K), and what would close the gap is to push identical builds and the identical probe through a path tracer sharing no code with ours. V-KID is a local-statistics descriptor, and PATCH-STITCH, built to exploit that, *outranks* the coherent but simple GABLED-HOUSE on both tiers, an ordering raters reverse unanimously (§6.6). That is the clearest limitation of the geometric tier, so PATCH-STITCH stays a standing control. V-KID is also blind to materials by construction, which is why the suite has two tiers and neither number belongs alone.

Our own protocol asks for $n \geq 256$ **and the leaderboard runs at** n=128, because the generator arms were sampled at 128; the encoder and view ablation and the LEGO replication are at n=256. We also score no published external model, so the leaderboard demonstrates what the suite does rather than re-ranking a published result. The one external comparison available without retraining is to a protocol rather than to a model: the generator released with this corpus (Chen et al., 2019) reports a cumulative chamfer distance, which is exactly invariant to our probe (§2).

7 CONCLUSION

Evaluation for 3D structure generation was inherited from image generation, and the inheritance carries a specific defect. Rendering a structure and embedding the renders measures its surface, and a building is not its surface. Filling every interior in a corpus of real buildings is invisible to a multi-view kernel distance that resolves every corruption reaching the exterior. That invisibility survives every encoder, view budget and statistic we try, and a cutaway view removes it. Adding the same mass to the outside of those buildings instead of the inside moves the render metric by a hundred noise floors, so what it misses is not the quantity of the change but its location.

A metric therefore earns the right to rank models by first resolving damage of known severity. The price is a handful of corruption arms scored beside the real reference in every run, and what it buys is that a result resting on a blind metric fails in the lab rather than in the literature. A metric inherits the blind spots of the observation it is built on. A protocol that sees only what a camera sees will rank models on the properties a camera can see, however strong the encoder behind it. Structure that decides whether a generated object is usable, and that no camera can photograph, has to be measured directly, and the same audit is owed to mesh and implicit 3D generation.

USE OF GENERATIVE AI

We used a generative AI coding assistant substantially in preparing this work, and describe the division of labor rather than a blanket disclosure.

Where it was used. Drafting and editing the text of this paper; implementing the figure-generation code; implementing parts of the evaluation suite, the probes and the human-study instrument; and conducting literature searches, including the prior-art check that identified Bińkowski et al. (2018)'s Appendix B.2 as the source of the mixture identity in Appendix L and the limitation already stated by Merino et al. (2026).

Where it was not. The research questions, the experimental design, the decision of what to measure, and the interpretation of results are the authors'. No experimental result, numerical value or table entry in this paper was produced by a language model: every number is emitted by the released code from a recorded run, and the figures read those runs directly rather than transcribing them.

Verification. Every citation was checked against arXiv or the official proceedings, and every numeral in the paper is re-derived from a recorded run by a registered check in the released code. The authors take full responsibility for the contents of the paper, including any remaining errors.

REPRODUCIBILITY STATEMENT

The evaluation suite, probes, procedural baselines, corpus split and human-study instrument are released. Every benchmark run writes a scorecard recording corpus, split, sample size, reference size, seed, host, device and commit, so each number in this paper is traceable to the run that produced it; Appendix C lists these. Two properties bear on replication in particular: controls are re-scored in every run rather than cached across runs, because the aggregate's unit is the spread of a control scored under the same protocol; and every arm must declare the split it was trained on, with arms that cannot reported as unranked (§6.5). **[TODO: Add the anonymized code URL and commit before submission, and insert the institutional determination covering the release. The licensing question itself is settled: Appendix Q states what we redistribute and what we do not, and the release contains no upstream content.]**

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A Corpus Construction and Provenance

Four public sources are pooled onto a shared legacy block vocabulary (the text2mc dump is remapped with 97% block-state coverage; world cuts are ground-stripped), passed through a quality gate, and deduplicated variant-aware. The fourth source, `text2mc_schem`, is the set of raw `.schem` files in the text2mc release that its authors never converted. It entered the pool after the 32^3 cache was built, so that cache draws on three sources and only the 48^3 cache carries all four. The main text (§5) describes the three that build `houses_32`, the corpus every reported run uses.

Drop reasons at 32^3, 826 in total: no interior 675, single-material blob 52, fragmented 37, too flat 29, fewer than three materials 22, too sparse 9, too few blocks 2.

Table 3: Corpus construction, transcribed from the cache manifests. Per-source columns are post-gate, post-deduplication counts and sum to **final**; **final** is *pooled* less *drops* less *dups*. Material variants, the same shape in different materials, are *kept* and not deduplicated.

cache	pooled	quality drops	exact dups	**final**	GrabCraft	3D-Craft	text2mc	text2mc _schem
houses_32	3,493	826	6	**2,661**	1,360	1,267	34	—
houses_48	4,502	1,381	33	**3,088**	1,410	1,381	180	117

The enclosed-air gate is calibrated per source. The gate requires at least 8 interior air voxels unreachable from the bounding box boundary. Measured on ground truth, 17% of GrabCraft houses legitimately have open interiors, against 31% of 3D-Craft and 54% of text2mc, so GrabCraft is exempted from the gate and the other three are not. `text2mc_schem` is gated on the same grounds as the rest of the text2mc release and is not calibrated separately. What the gate removes from the gated sources is trees, a truck and roof fragments, read off a pre-gate visual grid.

B Descriptor Ablation

Two objections apply to any handcrafted descriptor. The first is that V-KID separates Fill-Interior only because the probe adds mass, and the descriptor contains an explicit size term. The second is that it separates Fill-Interior only because the descriptor contains a sealed-interior scalar that the probe drives to zero. We test both by deleting the terms and by matching the mass.

Table 4 reports every variant at n=128 with 48 fresh real draws per variant, each in its own noise floor, so a column compares variants and not raw distances. Deleting the size term `log1p_n_blocks` moves the separation from 201.8 to 202.2. Deleting `interior_ratio_sealed` moves it to 197.1. Deleting both leaves 197.3. Neither term carries the result. No single block reproduces it alone either: the strongest, the nine global scalars, reaches 164.5 on 9 dimensions, and the 55 pattern bins that dominate the descriptor by dimension reach 113.9.

Grow-Shell settles the mass question directly. It dilates the solid outward until it has added exactly the number of blocks Fill-Interior adds to that build, which it achieves to within one block on 128 of 128 builds at a median of 196 added blocks. Same builds, same material, same mass, and the only difference is whether the new mass lands inside or outside. The two tiers then separate completely: MV-KID reads +0.26 noise floors on Fill-Interior and +102.27 on Grow-Shell, while V-KID reads 201.8 and 214.5. A descriptor that counted blocks could not produce that pair of rows.

C Reproducibility

[TODO: Fill in at submission: anonymized code URL, exact commit, the pinned split files, the feature-cache render canary hash, hardware, and per-run wall-clock. The suite writes a scorecard carrying corpus, split, n, reference size, seed, host, device and git SHA for every run, so this table is a transcription rather than a reconstruction.]

Two properties of the protocol matter for reproducibility:

- **Controls are re-scored in every run, never cached across runs.** BlockScore's unit is the spread of the real control *scored in the same run*, so two scores sit on one scale only when corpus, split, tier, seed, reference size and control size all match. We pin these as a named protocol and refuse to rank runs that do not satisfy it, listing each excluded run with its reason. The size of the control alone moves the score: the same model on the same corpus, split and tier scores 87.14 against a 128-build control and 19.05 against a 32-build one. A board that ignored the control's size would read that as a 4.5× improvement.
- **The calibration draw must be shuffled.** Taking the first n test builds as the real-vs-real draw, rather than a shuffled sample, inflates the measured noise floor by **10.4**× (s.d. 0.0318 against

Table 4: Descriptor ablation, n=128, 48 real draws per variant. Each separation is in that variant's own noise floor, measured under that variant. The two suspect terms are the explicit size scalar and the sealed-interior scalar; deleting either or both leaves the separation intact.

descriptor variant	dims	floor	FILL-INTERIOR	GROW-SHELL
full	77	0.1097	**201.8**	214.5
deleting a suspect term				
without `log1p_n_blocks`	76	0.1114	202.2	214.9
without `interior_ratio_sealed`	76	0.1127	197.1	211.7
without both	75	0.1147	197.3	211.9
deleting a whole block				
without pattern bins	22	0.5113	198.0	120.9
without wall thickness	72	0.1163	176.4	216.6
without height profile	69	0.1312	199.6	213.9
without global scalars	68	0.1302	171.6	206.7
one block alone				
pattern bins only	55	0.1802	113.9	201.3
wall thickness only	5	6.0167	158.7	75.7
height profile only	8	1.6293	8.0	28.9
global scalars only	9	1.7594	164.5	118.1

0.0030 at n=128) and shifts its center from $+0.0001$ to $+0.0404$. A contiguous block of a corpus stored in source and family order is not an exchangeable sample of it, so the floor it produces is both wider and displaced, and every separation measured against it is understated.

- **Every arm must declare its training split** to be ranked, and an arm that cannot is reported unranked with that as the stated reason. No arm in the run reported here declares one (§6.5). The requirement post-dates that run, so it is the fix rather than a gate that fired.

The language-model arms' decoding cap. LLM-FT and LLM-FT-GEO decode at temperature 0.8 under a 1,200-token cap. That cap is a real limitation of both arms: 11 of 128 generations came back empty, 139 attempts were needed for 128 samples, and several completions are truncated into structures one block thick (Figure 15). Raising the cap requires re-running the prompts, which we did not do for this submission.

D NEGATIVE RESULTS

Three experiments that did not work. Each bounds a claim a reader might otherwise expect us to make about injecting structure into sequence models.

Attention is not a free adjacency prior. A pretrained code LLM's self-attention over a serialized build recovers less adjacency information than the coordinates already printed in the prompt. Scoring all 336 heads by how well they rank the truly 6-adjacent previous pieces (AUC over the causal candidate set), the best head reaches 0.874 in the base model and 0.884 after task fine-tuning, well above a recency floor. But $-\|a_i - a_j\|_1$, three lines of arithmetic over the coordinates the serialization already prints on every line, scores 0.980. A trained bilinear probe does better, 0.914 at layer 14 of 28, and falls to 0.807 by the final layer, which is the layer generation actually reads. There is signal; there is nothing to steal.

A shared relative-3D attention bias is a null, for a reason in the parameterization. Adding a learned additive attention bias indexed by the relative 3D offset between pieces, zero-initialized and otherwise holding the fine-tune fixed, moved best validation loss by -0.0021 at one seed and did not touch the collapse mode. The learned table explains why: 364 of 729 offset buckets are *never updated*. Under raster emission order every earlier piece is sign-constrained, so half the offset space is unreachable by construction, and the arm's entire learned signal is a reparameterization of recency. Unlocking the rest requires a non-monotonic emission order. The reference anchor is not the constraint: the dead half of the offset space follows from monotonic emission alone.

E COUNTING INTERIORS

A naive enclosed-air count, restricted to air cells not reachable from the bounding-box boundary, counts a room with an open doorway as no room at all. We additionally close the solid under a 26-connected morphological closing that seals openings up to two blocks wide and report the air that becomes enclosed only then. Calibrated on a synthetic hollow cube with a 343-cell room: a 1×1, 1×2 or 2×2 doorway recovers all 343 cells; a 3×3 hole or a missing wall recovers 0, correctly classifying an open structure as having no room. On the same 200 held-out houses the probe statistics of §4.1 are measured on, $\mathbf{6.0}\%$ have no *sealed* interior at all, and *every one* of those twelve builds does have a room reachable through a doorway. Pooled over builds, apertures account for $\mathbf{25.6}\%$ of total real interior volume, which is the margin by which a sealed-only counter understates real houses. Every claim in the earlier literature, and in our own prior notes, that generated builds "have no interiors" was made with that counter.

The bootstrap p-value floor. A bootstrap sign-balance p-value is floored at $1/n_{\text{rep}}$, and Holm over the 55 pairs of an 11-arm run multiplies 0.005 to 0.275. Nothing can then reach $\alpha = 0.05$ however large the true difference, which put *every* arm of an earlier head-to-head, from held-out real to a monochrome corpus, in one "not separated" group. The reported p is now studentized and unfloored. The failure mode is silent: it produces a table in which nothing is significant, which reads like a careful result.

The dose axis is calibrated by construction. PARTIAL-FILL fills interior cells *inward from the walls*, nearest-first, which gives an exactly calibrated dose (interior ratio $0.1175 \to 0.0941 \to 0.0708 \to 0.0475 \to 0.0242$). Two earlier designs were rejected. Random interior speckle is detected as *noise* rather than as lost interior. Filling whole rooms largest-first makes the axis useless, because one dominant pocket is typically more than a quarter of the total interior, so a nominal 25% dose already removes 71% of it.

F DEFECTS FOUND IN OUR OWN GATES

These are the class of error a validation ladder exists to prevent, and it did not.

Two defects in our own gates. First, $\mathrm{sd}(\text{real})$, the unit every gate is stated in, was mismeasured twice. It was estimated from repeated subsamples of a pool that *excluded* the probe set, so the fresh real draw being tested was not a member of its own null. Overlapping draws also carry a finite-population factor that shrinks their spread. Corrected, a metric that separates the solidify probe at 67 s.d. passes the gate it had failed, having been rejected for noise that was measured wrong. Second, the n-stability gate tested a difference of means against the spread of a *single* draw, larger by $\sqrt{\text{reps}}$, and therefore had almost no power. Against the standard error of the difference, the biased estimator's $1/n$ decay is rejected at 14.8 standard errors.

A third defect, in the renderer. A feature extractor that fails open produces numbers that look excellent and cannot be caught downstream. Ours substituted a blank white frame for any rendering exception. Under a library version mismatch every render raised, every image came back white, every feature collapsed to one vector, and every arm scored KID $= -4 \times 10^{-13}$, better than real. The feature cache is validated by a render canary, so the all-white canary would have mismatched and the cache would have been silently rebuilt from blank frames. The renderer now raises past a 2% failure rate. Repairing the library reproduced the stored canary hash bit-for-bit, so no previously recorded number is affected.

G FULL TABLES

Tables reproduced here in full, including the results the main text reports only in prose.

Table 5 is the result. The solidified corpus reads 0.000 against a real floor of -0.001.

Significance, not interval overlap. Marginal confidence intervals are not a comparison, because non-overlap demands a difference roughly 41% larger than significance does. We therefore test

Table 5: Realism at n=128 against 399 held-out real builds, with bootstrap 95% intervals. **FILL-INTERIOR is indistinguishable from held-out real on the render metric** and 202 spreads away on the geometric one. The bottom two rows are the converse blindness: V-KID reads occupancy only, so a material permutation moves it by *exactly* zero (values are bit-identical to the real row, not merely close).

arm	MV-KID ↓	V-KID ↓
REAL (floor)	-0.001 $[-0.004, 0.003]$	-0.09 $[-0.22, 0.04]$
FILL-INTERIOR	$\mathbf{0.000}$ $[-0.002, 0.004]$	$\mathbf{22.03}$ $[20.65, 23.70]$
MATERIAL-SHUFFLE	0.674 $[0.644, 0.712]$	-0.09 *(bit-identical)*
MONOCHROME	0.930 $[0.903, 0.962]$	-0.09 *(bit-identical)*
TRAIN-COPY	0.015 $[0.007, 0.024]$	0.96 $[0.60, 1.51]$

Table 6: Dose-response, n=64 per rung. **range (sd)** is the best-to-worst span in units of the metric's own noise floor, given as n=64 / n=128 where both were run. The render metric is perfectly monotone on two axes and *flat* on the third; on that axis its rank correlation is unstable between the two runs ($+0.300$ vs $+0.900$), being the correlation of a sequence that does not move, so read the span instead. A resolution axis was dropped: changing a build's resolution changes its size, not its quality (§5).

axis	metric	Spearman	monotone	range (sd)
solidify (interiors filled, $0 \to 100\%$)	V-KID	$+1.000$	yes	$\mathbf{133 / 287}$
	MV-KID	$+0.300$ / $+0.900$	**no**	$\mathbf{0.4 / 1.1}$
occupancy noise ($0 \to 20\%$ blocks moved)	V-KID	$+1.000$	yes	226
	MV-KID	$+1.000$	yes	143
mixture (fraction of arm that is real)	V-KID	$+1.000$	yes	105
	MV-KID	$+1.000$	yes	121

every arm pair with the reference draw shared and apply Holm correction (Holm, 1979) across the family. Under that test FILL-INTERIOR and REAL **share a significance group** on MV-KID. The same test at the same n assigns TRAIN-COPY its own group at a difference of 0.015, a separation that sits on the α=0.05 boundary after Holm once the test is correctly calibrated (Appendix O), which is why the main text argues the point by magnitude instead. On V-KID the same solidified arm is four groups away, $\Delta = -22.13$ $[-24.08, -20.72]$, $p \approx 0$.

G.1 DOSE–RESPONSE

Table 6 gives every rung of the dose-response sweep, and Figure 10 plots the three axes. Table 7 gives the head-to-head letters for the case-study arms, and Figure 13 plots them with bootstrap intervals.

Every control is re-scored in every run rather than cached, because a distance means nothing until a floor has been measured under *this* run's corpus, split, n and reference size. Importing a floor across runs ranks arms against conditions they were never measured under.

What is not a control. Earlier versions of this suite included real data requantized to a coarser grid as a ranked control. We have removed it. Decimation changes a structure's *resolution*, not its *quality*, yielding a smaller building rather than a worse one, so a leaderboard row for it invites the comparison "this model is worse than 86%-deleted real data," which is not a meaningful statement about quality. It is removed from the validation ladder for the same reason and not merely from the leaderboard: a ladder rung asks whether a metric orders a *severity* axis correctly, and decimation is not one, so a metric that ranks it any particular way has demonstrated nothing. No result in this paper rests on a decimation rung.

Table 7: Retraining on a clean split collapses the margin. V-KID at n=128, 399-build reference, paired and Holm-corrected over all 153 pairs with an exact label-permutation test. Letters mark maximal sets of arms that no test separated at $\alpha = 0.05$: a shared letter means the pair was *not separated*, never that the arms are equivalent, and an arm carries several letters when non-separation fails to be transitive, as it does here. The contaminated arm is not separated from either its own clean retrain or the stronger language-model baseline, and the clean retrain ties with both baselines it had appeared to beat.

	arm	V-KID ↓	group	status
	REAL (floor)	−0.095	a	—
	TRAIN-COPY	0.960	b	DQ memorization
submissions	VOXEL-AR† *(contaminated)*	11.624	**d**	unranked
submissions	**VOXEL-AR** *(clean)*	14.898	**def**	ranked
submissions	LLM-FT	15.006	**de**	ranked
submissions	LLM-FT-GEO	15.869	**e**	ranked
submissions	VOXEL-AR-U *(clean)*	19.570	fgh	**DQ memorization**
	PATCH-STITCH	19.976	f	ranked
	RANDOM-VOXELS	37.729	j	ranked

Table 8: Every arm, with its full recipe. The compact version is Table 1 in §5.

kind	name	what it is
floor	REAL	held-out real builds from the test split, disjoint from the reference they are scored against
probe	FILL-INTERIOR	every enclosed air cell filled with the build's own dominant material
probe	MATERIAL-SHUFFLE	the block multiset permuted across the build's own occupied cells: geometry exactly preserved
probe	MONOCHROME	every block replaced by one material: palette destroyed, geometry exactly preserved
probe	TRAIN-COPY	verbatim copies of training builds
probe	SINGLE-MODE	one build repeated n times
baseline	RANDOM-VOXELS	random voxels at the real occupancy density
baseline	HOLLOW-BOX	a hollow box: floor, walls, flat roof
baseline	GABLED-HOUSE	box, pitched roof, door, windows: a competent afternoon of rules
baseline	PATCH-STITCH	8^3 patches lifted verbatim from *different* real builds
baseline	TRAIN-COPY+NOISE	a training build with 10% of its blocks displaced
submission	VOXEL-AR	autoregressive transformer over merged-piece tokens, trained on the canonical split
submission	VOXEL-AR-U	the same model without orientation augmentation
submission	LLM-FT	a fine-tuned code language model emitting one line per piece
submission	LLM-FT-GEO	LLM-FT with a learned relative-3D attention bias

G.2 METRICS

We use two realism metrics throughout, both the unbiased kernel two-sample statistic of Bińkowski et al. (2018) and differing only in the features they are computed on.

- **MV-KID**, the *multi-view kernel distance*. Each structure is rendered from four orthographic views at 224px and embedded with DINOv2-B (Oquab et al., 2024). This is the render-and-embed family the field uses, and is our stand-in for it.
- **V-KID**, the *voxel kernel distance*. The same estimator on a 77-dimensional descriptor computed directly on the voxel grid, with no rendering and no GPU. The dimensions are four blocks: the distribution over the **55** D4-orbits of $2 \times 2 \times 2$ occupancy patterns (256 patterns fall into 55 orbits by Burnside's lemma under the order-8 group of yaw rotations and an x-mirror), a **5**-bin wall-

Table 9: The render tier's metric-validation ladder (DINOv2-B, 4 orthographic views at 224px, n=64). Columns are the probe rungs: real held-out, retype 10% of blocks (the rung the ordering gate uses for this tier), displace 10%, delete 40%, material shuffle and a 90° rotation; **sd** is the metric's own noise floor. Gates are blocking, and *only **KID** and **MMD-RBF** pass*: both PRDC recall and coverage fail n-stability. MMD-RBF is on its own scale and is reported to one decimal; the last row is the biased estimator.

metric	real	retype 10%	displace 10%	del. 40%	mat. shuffle	rot. 90°	noise floor	gate
KID (unbiased)	**0.013**	**0.178**	**0.447**	**0.108**	**0.782**	**0.011**	**0.008**	**PASS**
MMD-RBF	**3.8**	**53.5**	**141.7**	**32.9**	**226.5**	**3.3**	**2.5**	**PASS**
1−recall	0.113	0.315	0.919	0.323	0.750	0.121	0.027	FAIL: n-stab.
Fréchet distance	0.107	0.161	0.252	0.147	0.374	0.105	0.005	FAIL: n-stab.
1−coverage	0.258	0.512	0.839	0.516	0.952	0.282	0.052	FAIL: n-stab.
1−density	0.000	0.259	0.594	0.216	0.881	0.000	0.071	FAIL: noise ord.
1−precision	0.125	0.188	0.203	0.141	0.531	0.109	0.043	FAIL: noise ord.
biased MMD estimator	0.082	0.502	1.178	0.321	2.037	0.078	0.022	FAIL: n-stab.

thickness histogram, an **8**-bin height profile normalized by the build's own height, and **9** global scalars. The group is exactly D4: a vertical flip is deliberately *not* quotiented out, since upside-down is a real defect in a building, and we confirm it moves 45 of the 77 dimensions. Patterns are counted over the volume padded by one air cell, so surface patterns are counted rather than clipped at the border. The all-empty orbit is identically zero after normalization, leaving 76 informative dimensions and, since each of the three histograms sums to one, 73 free parameters.

Whitening and clipping. Descriptor blocks differ by orders of magnitude in scale, so the feature map whitens against statistics frozen on the reference, **clips at ± 5 reference standard deviations**, and then L2-normalizes each row. The clip is load-bearing rather than cosmetic and we report it for that reason: real houses have almost no mass five voxels deep, so the reference standard deviation of the thickness ≥ 5 bin is 1.6×10^{-4}, and FILL-INTERIOR moves that single bin by 67.5 reference standard deviations before clipping. Without the clip one coordinate would dominate the kernel, and the separation we report would be a one-dimensional thickness test wearing a 77-dimensional coat.

Both are reported in *noise floors*: the spread of repeated real-vs-reference draws at the same n. A value of 1 means one sampling-noise unit away from real, which makes the two commensurable despite differing by three orders of magnitude in raw units.

G.3 METRIC VALIDATION

Table 10 reports the procedural baselines against real held-out data, Figure 11 plots Table 11, Figure 9 shows both probe directions on three builds, Figure 12 plots the two pillars against each other, and Figure 14 walks through the geometric descriptor.

Every candidate metric is scored on rungs of known damage, in units of its own noise floor (the spread of real-vs-reference draws at the same n). Validity gates are *blocking*: a metric that fails one is reported as *gated* with the failing gate named, and may not appear in any ranking. Sensitivity, how *much* damage a metric resolves, is recorded separately, because that is a question about n, not about whether the metric measures the right thing.

Why coherence is reported as distance to real. Two rows at the extremes of Table 2 show why the suite reports more than its two realism pillars. An arm of *verbatim copies of the training set* scores 0.015 raw MV-KID, third of sixteen arms, behind only held-out real data and FILL-INTERIOR, and only the novelty columns catch it: nearest-neighbor IoU 1.000, duplicate rate 1.000 and a retrieval percentile of 0.000 against a real held-out floor of 0.540. At the other end of the scale the procedural baselines are *too* hollow, with aperture-aware interior ratios of 0.61 (HOLLOW-BOX) and 0.56 (GABLED-HOUSE) against a real 0.13. A coherence measure read as "higher is better" would therefore rank a hollow box above a house, which is why every coherence measure is reported as distance to real.

Table 10: Procedural baselines, with real held-out data for reference. GABLED-HOUSE is the bar that matters: what a competent afternoon of hand-written rules produces. PATCH-STITCH is an attack, not a baseline. Every cell is a per-build mean over $n{=}128$ builds per baseline; the real row is all 399 held-out test builds. **Interior is reported twice**, because the two counters answer different questions (Appendix E): *sealed* is air unreachable from the bounding box, *open* additionally credits rooms reachable only through a door or window. The distinction is not cosmetic: GABLED-HOUSE has a doorway, so it scores 0.000 sealed and 0.562 open.

				interior		
baseline	what it is	blocks	wall frac	sealed	open	LCC
RANDOM-VOXELS	random voxels at real density	1383	0.049	0.002	0.003	0.406
HOLLOW-BOX	hollow box, floor, walls, flat roof	1345	1.000	0.609	0.609	1.000
PATCH-STITCH	8^3 patches from *different* builds	1405	0.624	0.003	0.009	0.866
GABLED-HOUSE	box, pitched roof, door, windows	1139	0.969	0.000	0.562	1.000
TRAIN-COPY+NOISE	a training build, 10% displaced	1057	0.454	0.002	0.055	0.967
real	held-out test split	924	0.745	0.114	0.129	0.978

H HUMAN STUDY PROTOCOL

Protocol. Nine arms give 36 pairs; 288 trials, 40 per session, sides randomized per trial, and arm labels stripped from the page, all enforced by an automated check, not merely intended, because a condition visible in the DOM is a condition the rater can read. Catch trials (real vs. uniform noise) are interleaved.

Analysis, fixed in advance and in this order.

1. Exclude any session scoring below 75% on catch trials. A rater who cannot distinguish a real house from uniform noise is not doing the task, and the threshold is fixed here rather than tuned after seeing which exclusion helps a metric.
2. Fit **Bradley–Terry** arm strengths (Bradley & Terry, 1952) rather than raw win rate. Win rate depends on which opponents an arm happened to be drawn against, and with a random subset shown per session, coverage is always uneven.
3. Report (a) pairwise agreement on humanly-decided pairs and (b) Spearman ρ against the Bradley–Terry scores. Pairs where humans are near chance are reported but excluded from the agreement statistic: counting them scores a metric on coin flips, and with enough of them any metric converges to 50%.

The pipeline is validated end to end on 14 simulated rater sessions, two of them answering at random. It excluded exactly those two on catch trials and recovered the injected quality ordering.

I EXTENDED RELATED WORK

O'Bray et al. (2022). Perturbs graphs at graded severity and scores a metric on whether it responds monotonically, with Pearson correlation against perturbation degree as the pass criterion. The procedure selects among metric hyperparameters; it never disqualifies a metric, has no invariance axis, and does not test stability in the sample size. Our ladder is a blocking precondition on ranking and adds both.

Borji (2019). Enumerates desiderata for generative metrics including sensitivity to distortions, invariance to semantically-meaningless transforms, and low sample complexity, which are our ordering, invariance and n-stability gates respectively. The contribution here is operationalizing them as a gate that runs in every evaluation rather than as a list.

Caytuiro & Sipiran (2025). Audits 3D generators for symmetry violation, with the same rhetorical structure as this paper one property over. The contrast is the point: a broken symmetry is visible

in any exterior view, so a render-and-embed metric can in principle see it. An interior cannot be seen from outside at all, which is why the failure is structural rather than a matter of encoder quality.

Luan et al. (2026). Criticizes rendering-based 3D evaluation on the grounds of incomplete structural coverage and viewpoint sensitivity, and replaces renders with sampled surface points. This removes rendering without removing the blindness: a surface point set describes the same boundary. We read this as support: two independent critiques of the render pipeline, neither of whose fixes reaches the interior.

Hervé & Salge (2021). Correlates eighteen procedural-content metrics against GDMC jury scores. The only prior meta-evaluation of metrics in this domain, and the precedent for validating a metric against human judgment here rather than assuming it transfers.

J Full View-Protocol Ablation

Figure 4 plots every condition in Table 12 on one axis.

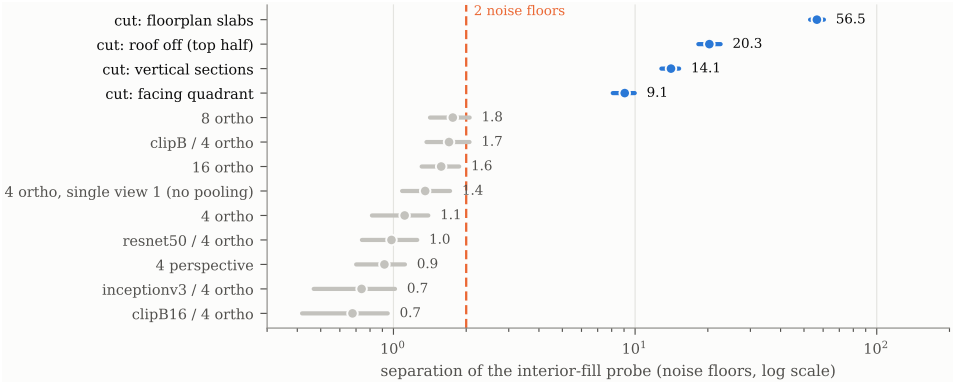


Figure 4: **The blindness belongs to the view protocol, not the encoder.** Separation of the interior-fill probe in each condition's own noise floors (log scale), with paired bootstrap intervals. Gray: every encoder, view budget and projection we test on exterior views, spanning 0.7–1.8 floors. Blue: the same encoder and the same statistic once the renders expose the interior.

Table 11: The in-domain ladder at n=128, each statistic in its own noise floors, measured from the floor's mean rather than from zero, so these values are not directly comparable with Table 12's. **All seven miss Fill-Interior and resolve every other corruption.**

statistic	Fill-Interior	20% deleted	Material-Shuffle
KID	**2.0**	12.3	132.3
MMD-RBF	**2.1**	13.9	148.8
Fréchet distance	**0.5**	10.8	72.4
1−coverage	**0.2**	7.7	19.8
1−density	**0.1**	7.2	14.0
1−precision	**−0.4**	9.0	28.0
1−recall	**0.4**	1.4	30.7

K What a Render Protocol Provably Cannot See

A deterministic render pipeline R_V with a fixed view set V induces an equivalence relation on structures: $x \sim x'$ exactly when $R_V(x) = R_V(x')$.

Lemma 1 (Render kernel)**.** *If $x \sim x'$ then $T(\mathcal{E}(R_V(x))) = T(\mathcal{E}(R_V(x')))$ for every encoder $\mathcal{E}$, every pooling rule and every statistic T.*

Table 12: Separation of FILL-INTERIOR from real in each condition's own noise floors (n=256, paired bootstrap, 95% intervals). **Every exterior condition lands between 0.7 and 1.8 noise floors; cutaways reach 9–57 with the same encoder and the same statistic.**

	condition	separation	sees interior?
encoder	DINOv2-B (ours)	+1.1 [0.8, 1.4]	no
	CLIP ViT-B/16	+0.7 [0.4, 0.9]	no
	CLIP ViT-B/32	+1.7 [1.4, 2.1]	no
	ResNet-50 (ImageNet-supervised)	+1.0 [0.7, 1.3]	no
	Inception-v3 pool3 (the FID backbone)	+0.7 [0.5, 1.0]	no
views	8 orthographic	+1.8 [1.4, 2.1]	no
	16 orthographic	+1.6 [1.3, 1.9]	no
	perspective	+0.9 [0.7, 1.1]	no
cut	**facing quadrant removed**	**+9.1** [8.1, 10.0]	**yes**
	roof removed	**+20.3** [18.3, 22.6]	**yes**
	floorplan cross-sections	**+56.5** [53.0, 60.5]	**yes**
	V-KID (no rendering at all)	+120.8 [115.7, 126.3]	yes

Proof. Immediate: the composition is a function of $R_V(x)$ alone. □

On its own Lemma 1 is a tautology. It becomes a statement about this paper because FILL-INTERIOR lands inside an equivalence class, which needs two further facts.

Lemma 2 (Surface invariance)**.** FILL-INTERIOR *leaves the exterior-air region unchanged, and leaves every exterior face unchanged in cell, direction and material.*

Lemma 3 (Occlusion)**.** *Assume (a) nearest-surface shading, (b) faces bordering exterior air are opaque, and (c) every camera lies outside the bounding box. Then* $R_V(\text{FILL-INTERIOR}(x)) = R_V(x)$ *for* every *view set* V.

Proof sketch. Any ray reaching a newly filled cell must first cross a face bordering exterior air. By (b) that face is opaque and by (a) it is what the ray returns, so no pixel can depend on the filled interior. □

Lemma 2 has a consequence beyond rendering. Any metric computed on samples of the *exterior surface* is a function of that face set, so it too is constant under FILL-INTERIOR. We verify it: sampling exterior face centers from a build and from its solidified copy gives **identical point sets on 128 of 128 builds**, so chamfer distance, earth-mover distance and every other point-cloud distance between them is exactly zero. This matters for the proposal to replace renders with surface samples (Luan et al., 2026), and for this corpus in particular: the model released with it (Chen et al., 2019) reports a cumulative chamfer distance. Sampling *all* exposed faces, including those bordering sealed cavities, does see the probe (mean chamfer 0.485), but a surface sampler that reached interior cavity walls would be observing the interior, which is the thing at issue.

Together with Lemma 1 this is an impossibility result, not a statistical one: under an opaque renderer *no* encoder, statistic, sample size or view budget separates a corpus from its solidified copy. We verify Lemma 2 exhaustively (0 violations over the 399 held-out builds, and 0 for HOLLOW-SHELL) and Lemma 3 by rendering, using counterfactual texture libraries to switch each hypothesis off in turn. The four rows below are one experiment over the same 1,024 (build, view) pairs, so they are comparable to each other; the 91% and 1.75% quoted in §6.3 come from a separate sweep of the whole test split (399 builds, 1,596 views), which is why the deployed row here reads 1.67% rather than 1.75%. The two agree to within the difference between a subsample and a census, and we keep each number with its own experiment rather than mixing them:

renderer	bit-identical	px changed (med.)	max ch. (med.)
opaque, flat tiles (Lemma 3)	**100.0%**	—	—
flat tiles, translucency on	62.5%	0.80%	94
textured, translucency off	8.1%	0.99%	78
deployed (textured, translucency on)	8.6%	1.67%	211

Our renderer violates hypothesis (b) in two ways, and the two have different standing. The first is physical: Minecraft glass is translucent, so a window in front of a filled void reveals the fill. This is a real channel, and it is why the deployed figure is 91% rather than 0%. The second is an artifact of our mesh builder, which we report because it inflates that count. `build_mesh` assigns texture-atlas tiles in face-iteration order over *all* exposed faces, interior ones included; deleting interior faces changes the tile count, hence the atlas layout, hence every UV rectangle, and the sampler is nearest-neighbor. Switching translucency off still leaves 92% of views altered by this alone, and within that condition "atlas unchanged" predicts bit-identity at 98.7% agreement. A reader should therefore take 91% as a property of our renderer, not of the corruption; the corruption's own reach is the 28% that survives flattening the textures.

We then rescored the probe end to end through each of the four pipelines, holding the arms, the draw and the estimator fixed and changing only the texture pack. This turns the lemma into a prediction the benchmark either satisfies or does not:

renderer	MV-KID separation	features identical
opaque, flat tiles	$+\mathbf{0.000}$	**yes**
flat tiles, translucency on	-0.069	no
textured, translucency off	$+0.026$	no
deployed	$+0.119$	no

The prediction holds exactly: bit-identical renders give bit-identical features and therefore an identical kernel distance, so the separation is 0 as an identity rather than as a rounded measurement. The other three rows carry the more useful lesson. The residual the benchmark reports for this probe is not a small signal. It is *wobble*. Changing only the texture pack, which changes nothing whatever about the geometry being probed, moves it from $+0.26$ to $+0.03$ to -0.16: a range wider than the value itself, and one that crosses zero, with the probe reading *closer* to real than real does. This is what it looks like when a metric is flat on an axis rather than weakly sensitive to it, and it is why we argue the point by magnitude and ordering rather than by the residual's sign. (The deployed row reads $+0.26$ here against the leaderboard's $+0.38$ for the same quantity; the two runs estimate their own floors, 0.00358 and 0.00306, which is the effect Appendix N documents.)

Where the surviving signal dies is measurable but not, we think, provable. In DINOv2 the patch tokens a changed pixel touches move by 0.062 against a between-build scale of 0.517 (12%); the CLS token moves 0.011 against 0.277 (3.9%); the four-view pooled embedding moves 0.0053 against 0.268 (2.0%). The signal enters the features and is attenuated by aggregation. We report this as a measurement and decline the tempting story that the encoder is invariant to the corruption by training, which the patch-level numbers contradict.

The converse holds by the same style of argument and makes the paper's complementarity claim provable on both sides rather than measured on one.

Lemma 4 (Material blindness of V-KID)**.** *The descriptor of §G.2 factors through the occupancy mask, so any transformation preserving occupancy,* MATERIAL-SHUFFLE *and* MONOCHROME *among them, leaves it pointwise unchanged, and V-KID returns the real arm's value exactly.*

Verified: descriptors are bit-identical (max difference 0.0) and V-KID returns the identical float for all three arms. The $+0.00$ printed in Table 2 is an identity, not a rounded measurement. Renders of those same builds are never identical (0 of 256 views; a median 25% of pixels move), so material permutation provably escapes the render kernel exactly as interior fill provably falls inside it.

L READING A SQUARED STATISTIC OVER A POPULATION OF PROMPTS

Everything in the main text concerns what a metric can *see*. How its value is *read* is a second issue, and it affects any leaderboard reporting KID or MMD over a population of prompts. Because these statistics are squared, an arm that fails on a quarter of its builds scores only 6.4% of the way to fully bad, where a reader assuming proportionality would infer a quarter of the damage. Reporting the square root recovers the failure rate to a mean absolute error of 0.013 against 0.128 for the raw value.

A generator fails on a subset of prompts rather than uniformly. For a squared-MMD statistic the response to such a mixture is quadratic,

$$\mathrm{MMD}^2(\alpha P + (1-\alpha)Q,\ P) = (1-\alpha)^2\,\mathrm{MMD}^2(Q, P), \tag{2}$$

a one-line consequence of the linearity of the mean embedding. Bińkowski et al. (2018) state this identity in their Appendix B.2, and both they and Heusel et al. (2017) plot their metrics against a contamination fraction. We add the reporting recommendation it implies: report $\mathrm{MMD} = \sqrt{\mathrm{MMD}^2}$, which is *linear* in the contaminated fraction and free to compute. The identity holds for squared integral probability metrics such as KID and MMD. It does not hold for Fréchet distance, which Bińkowski et al. (2018) show is not polynomial in α, so we derive the correction for KID only.

Measured on a mixture ladder whose bad pool is RANDOM-VOXELS at $n{=}128$, the best-fit exponent is $\mathbf{1.99}$ against the identity's 2. An arm failing on a quarter of its builds scores 6.4% of the way to fully bad (0.110 against 1.718). Reporting $\sqrt{\mathrm{KID}}/\sqrt{\mathrm{KID}_{\mathrm{worst}}}$ recovers the true bad fraction to a mean absolute error of $\mathbf{0.011}$ (MV-KID) and $\mathbf{0.022}$ (V-KID), against 0.124 and 0.135 for the raw values.

One caution on the readout. The identity holds for the population quantity, while we report an unbiased estimator of MMD^2, which can be negative and whose square root is therefore biased downward by Jensen's inequality, worst where the statistic is small relative to its own standard error. We report the square root as a diagnostic for reading a failure rate, never as the ranked quantity.

M WHAT BLOCKSCORE'S MAXIMUM IS, STATISTICALLY

BlockScore is the maximum of five pillar scores, and a maximum of several mean-zero quantities is not itself mean-zero. The control arm reads exactly 0.00 on every pillar because it *is* the arm the calibration is taken from: $(v_p - v_p)/\mathrm{sd}_p = 0$ identically, with zero variance. That is an identity, not a measurement, and Table 2 labels it as such. A *fresh* real arm, drawn from the same distribution but not used to calibrate, scores $\mathbb{E}[\text{BlockScore}] = 1.08$ with standard deviation 0.89, right-skewed ($+0.89$), and a 95th percentile of 2.8. Read BlockScore against that null rather than against zero: a value of 15 is not "15 standard deviations" but "far outside a null whose upper tail is under 3". The inflation is about one noise floor, against a smallest ranked score of 60.8, so it changes no ordering and no conclusion in this paper. A future suite comparing arms that sit within a few floors of real would need to subtract it.

The five pillars are not in one unit. Appearance, geometry and palette are divided by a genuine sampling noise floor, and we verify this: their declared units match the spread of fresh real draws to within 7% (0.93, 1.03 and 1.05 of the measured value). The coherence and geometry-scalar pillars are divided by a per-build robust spread of the reference, an effect size measuring how far apart two distributions are in units of how much individual builds differ Measurement puts that unit $29\times$ and $52\times$ coarser than those pillars' own sampling noise. A structure pillar would therefore have to move some $1{,}800$ of its actual noise floors to veto a live pillar sitting at $+60$, and on our card none ever does. We state this as a limitation of the current calibration rather than a property of the aggregation rule: the maximum is the right rule, and two of the five inputs are quoted in a unit that makes them diagnostics rather than vetoes.

One consequence for reporting. Because the winning pillar's unit is what the number is in, BlockScore is uninterpretable without the pillar that produced it, which is why every leaderboard we print carries a *worst pillar* column. It is also not convertible to a p-value: the null is max-shaped and skewed, so the usual normal reading does not apply.

N THE NOISE-FLOOR UNIT DEPENDS ON n

Separations throughout this paper are quoted in *noise floors*: the standard deviation of repeated real-versus-reference draws at the same sample size. That unit makes two metrics with incommensurable raw scales comparable within a run, which is why we use it. It is not scale-free across runs, and we state the dependence explicitly because it is easy to misread.

Measured on the identical corruption at two sample sizes:

metric	sd n=64	sd n=128	ratio	sep n=64	sep n=128	ratio
V-KID	0.1732	0.0848	**0.49**	132.9	286.7	**2.16**
MV-KID	0.00449	0.00331	0.74	0.4	1.1	2.84

The noise floor of the geometric metric halves when n doubles, the $1/n$ decay expected of a degenerate U-statistic under the null, while the population effect is fixed. A separation in noise floors therefore grows *linearly* in n, and a figure such as "202" means "202 at n=128"; the same measurement at n=256 reads roughly 400.

Three practical consequences. Separations are only comparable across runs at equal n, which is why our protocol pins it. Cross-domain figures must be normalized before they are set side by side. And regenerating every arm at a larger n leaves the *ordering* of arms intact while roughly doubling every magnitude, and the orderings are what our conclusions rest on.

The unit is estimated, not only n-dependent, and equal n is not enough. The floor is itself a statistic: 48 real-versus-reference draws, carrying a relative standard error near $1/\sqrt{2 \cdot 47} \approx 10\%$. Two of our own runs make the consequence concrete. At the same n=128, the same corpus, the same pinned split, the same reference and the same commit, the leaderboard run of record reports the interior-fill separation as $\mathbf{201.8}$ and the converse-probe run reports it as $\mathbf{286.7}$. The only differing inputs are two random seeds, one choosing which 128 test builds were drawn and one seeding the 48 floor draws. The ratio decomposes exactly into the two:

$$\underbrace{\frac{286.7}{201.8}}_{1.42} = \underbrace{\frac{0.1097}{0.0848}}_{\text{floor estimate, } 1.29} \times \underbrace{\frac{24.430 - 0.128}{22.034 - (-0.095)}}_{\text{raw difference, } 1.10},$$

that is, roughly 70% a different estimate of the same floor and 30% a different draw of the builds. Two estimates of one floor landing 29% apart is a 2σ event for a standard deviation taken from 48 draws, so the unit carries a visible error bar. A separation should therefore be read to one or two significant figures, with its run named. We quote 202 throughout because it is the run of record's value, and the reader should understand it as *about two hundred*: a sweep over twelve seeds places it between 185 and 274.

The linear-in-n rule holds at fixed reference size. The prediction above, that n=256 reads roughly 400, assumes the reference pool is unchanged. Our view-protocol ablation measures the same probe on the same corpus and split at n=256 and reads 120.8, not 400, because it scores against 512 reference builds rather than 399 and its floor (0.188) is consequently *larger* than the n=128 floor rather than half of it. The rule describes how a separation moves when n grows within one protocol; it does not survive a change of reference. This is the same caution as the cross-domain one, and it is why every number in this paper is reported with its run.

A note on what the ladder tables difference against. The per-metric ladder artifacts (Table 9) express a rung as its distance from the null *mean*, while the leaderboard expresses an arm as its distance from a held-out REAL *draw*. On an exactly invariant rung the two conventions differ by construction: the geometry metric's material-shuffle row is 0 exactly against the REAL draw and 1.19 against the null mean. The conventions must not be mixed within one comparison.

MV-KID scales less cleanly (0.74 rather than 0.49) because its separation is a ratio of two quantities that are both near zero. That is the finding rather than a defect of the unit: the render row should be read as "indistinguishable", not as a number.

O CALIBRATION OF THE HEAD-TO-HEAD TEST

Our first implementation estimated the standard error of a paired difference by subsampling 80% of each arm without replacement and using the spread of those draws directly. That is the variability of a subsample statistic around the full-sample value, which becomes the standard error of the full-sample statistic only after a rescale we did not apply. Measured on exact nulls, the resulting test rejected 3.4× too often at α=0.05 when the arms match the reference and 7.2× too often when they do not.

No constant rescale repairs it. Conditional on a fixed reference, the difference of kernel distances is a degenerate U-statistic when both arms match the reference and a non-degenerate one when they do not, with required factors of 1.31 and 2.06 respectively; across the 153 pairs of one table the factor ranges from 0.30 to 11.9. We therefore use an exact label-permutation test: pool the two arms, re-split at the observed sizes, and recompute the difference against the same fixed reference. Its null standard deviation tracks a Monte-Carlo ground truth within 5% in both regimes, and its type-I error is nominal at both levels.

Recomputing the run of record changes only claims of separation, which is the expected direction: a test that rejects too easily can manufacture a difference but never a tie. Both of our null results hold and strengthen: FILL-INTERIOR remains unseparated from held-out real (p from 0.61 to 0.77), and the clean retrain remains unseparated from the language-model baseline (0.89 to 0.93). Two positive separations do not survive: TRAIN-COPY against held-out real falls to the α=0.05 boundary, and the contaminated arm separates from one language-model baseline but not the other. We report the blindness result by magnitude for this reason, and the case study's conclusion, that the contaminated arm's apparent advantage does not survive a clean split, is unaffected, since it rests on a within-architecture comparison.

P ADDITIONAL FIGURES

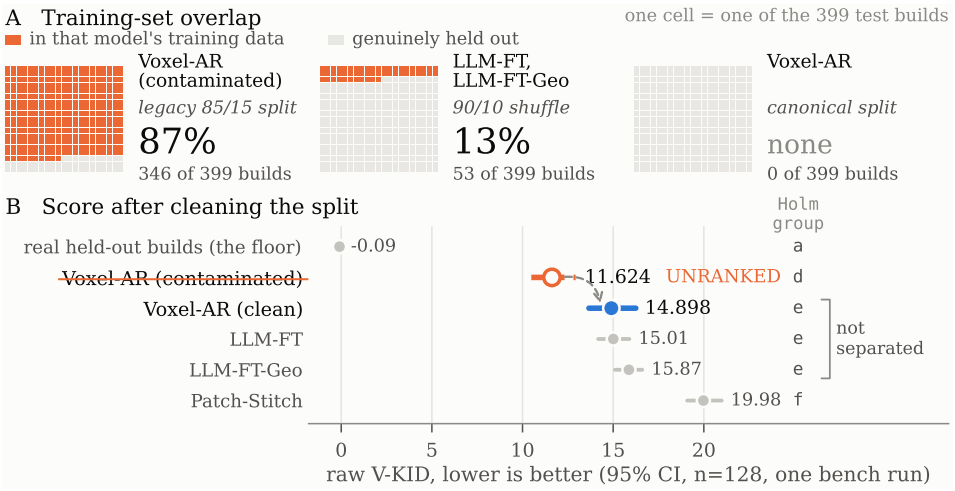


Figure 5: **Declared provenance, and what it cost.** *A:* how much of the canonical test split each arm had already trained on; the third waffle is VOXEL-AR retrained on that split. *B:* the retrain costs +3.27 raw V-KID and erases the margin. No memorization gate fires on the contaminated arm, so it is unranked rather than disqualified.

Q DATA PROVENANCE AND RELEASE

Our Minecraft corpus derives from four public sources: the 3D-Craft house dataset (Chen et al., 2019; Gray et al., 2019), released with the CraftAssist project; a scrape of `grabcraft.com`; and the text2mc dump, whose builds originate from `planetminecraft.com`. **We do not redistribute the GrabCraft- or text2mc-derived structures.** GrabCraft's terms permit sharing

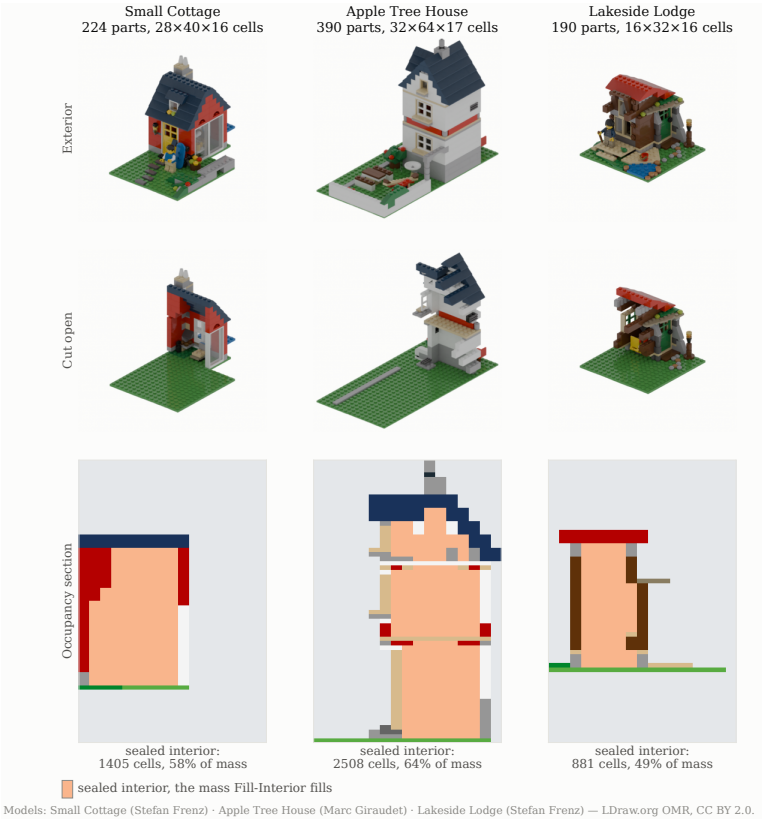


Figure 6: **A second domain, with real colors.** Three LDraw.org OMR models in their own LEGO colors: the exterior an exterior protocol sees, the same parts cut past one plane, and the occupancy grid we derive sliced at that plane, sealed interior in peach, every other color the parts' own, all renders at one scale. This domain shares only the fact that builds enclose space: different parts, different lattice, different renderer, 190–390 parts against thousands of blocks. The measured replication of §6.3 runs on StableText2Brick rather than on these three models, which we picture because their terms permit it (Appendix Q). Official LEGO sets are far less hollow than either measured corpus: among models with at least eight sealed cells, the basis the OMR audit reports, FILL-INTERIOR adds a median $+1.2\%$ mass to an OMR model against $+35.8\%$ to a held-out house. Models: Small Cottage (Stefan Frenz), Apple Tree House (Marc Giraudet), Lakeside Lodge (Stefan Frenz); LDraw.org OMR, CC BY 2.0, modified only by the deletions shown.

blueprints only by linking, and PlanetMinecraft's terms prohibit redistributing site data. We release instead the curation code, the per-build source URLs, content hashes, the group-aware split as index lists, and a rebuild script, so the corpus reproduces bit-exactly without our distributing third-party content. The four things the paper promises, namely the suite, the probes, the baselines and the split, contain no upstream content.

Every build shown in this paper, and every stimulus shown to a human rater, comes from **3D-Craft only**. GrabCraft's terms name pictures of blueprints as a prohibited form of distribution; 3D-Craft's builds were commissioned as a research corpus and carry no prohibition on depiction. Their license needs stating precisely, because it is not the one the CraftAssist code carries. The CraftAssist codebase is MIT, but the house data ships as a tarball with no license file, and the release that distributes it for the 3D-Craft paper, `facebookresearch/voxelcnn`, is CC BY-NC 4.0. We therefore treat the corpus as non-commercial research material requiring attribution, which is what our use and this paper's figures are, and we state that restriction on the release rather than inheriting the code's MIT terms. Figures render these builds with Minecraft's own texture assets, which are Mojang's copyright. Mojang's usage guidelines permit non-commercial use of images of the game, and this paper is non-commercial research; we also reproduce the attribution they

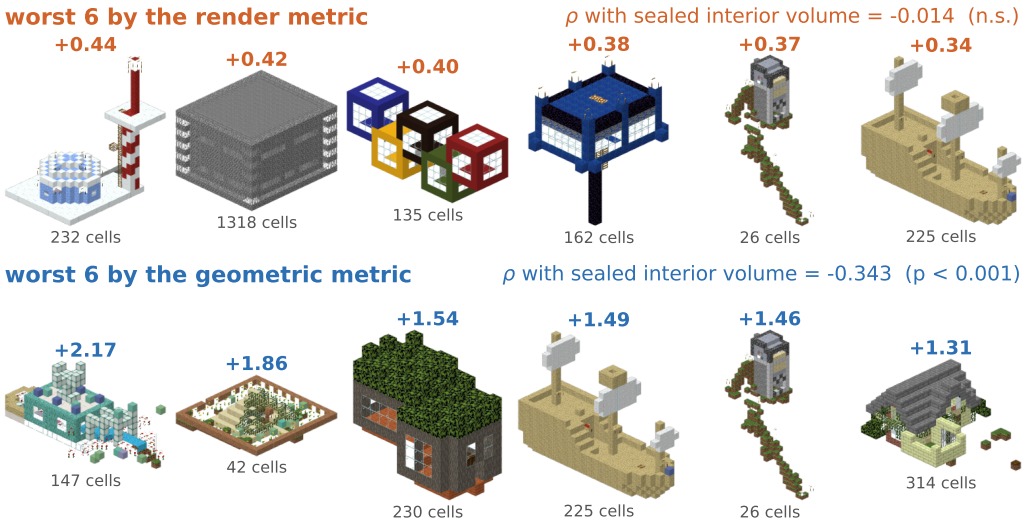


Figure 7: **The two metrics disagree about real builds, with no probe involved.** The unbiased estimator is a sum over builds, so each build's contribution to its arm's score is recoverable. The six builds each metric penalizes most are shown with their contribution and their sealed interior volume. The two orderings are nearly independent, $\rho = +0.167$, and only the geometric ordering tracks how much interior a build has: its rank correlation with sealed interior volume is -0.343 ($p < 0.001$) against -0.014 for the render metric. Builds shown are 3D-Craft only, for licensing.

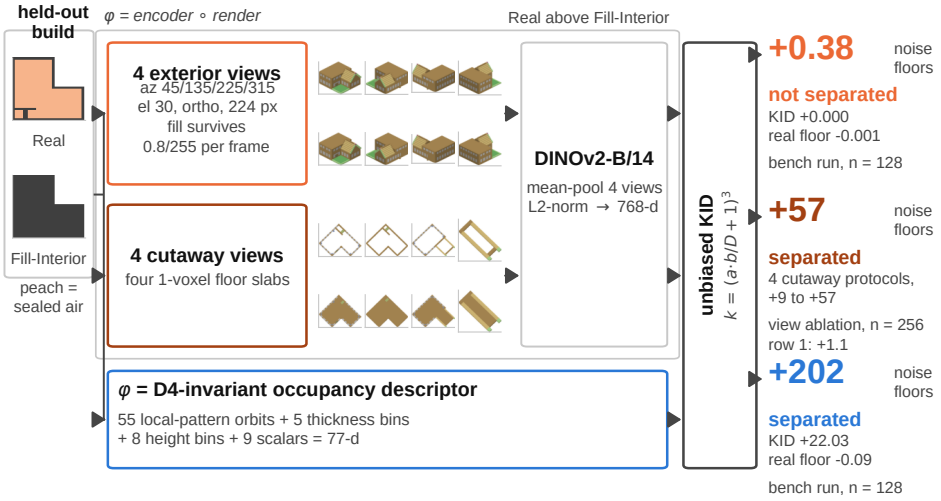


Figure 8: **One estimator, three feature maps.** MV-KID and V-KID are the same unbiased kernel distance of Eq. 1 and differ only in ϕ. *Row 1:* the render-and-embed path, where the structure reaches the statistic as four images. *Row 2:* the same renderer and the same encoder with cutaway views, which recovers the signal. One renderer, one encoder, only the views differ, and four other encoders and 8 or 16 views change nothing (Table 12). *Row 3:* the geometric path, which reads occupancy directly and needs no renderer and no GPU. A noise floor is a run's own real-vs-real spread and falls with n, so separations compare within a run: rows 1 and 3 come from one scorecard at n=128 against 399 reference builds, and row 2 from the view ablation at n=256, where row 1 scores $+1.1$. The sections at left are horizontal slices of one 3D-Craft build.

require, and nothing here suggests official endorsement. Two limits follow and we observe both.

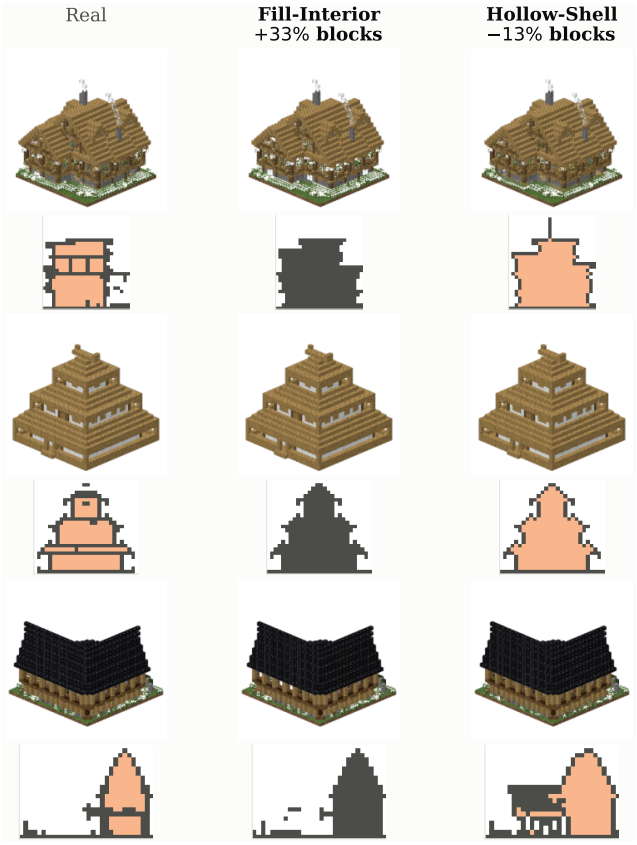


Figure 9: **Both directions of the corruption, on three held-out builds.** FILL-INTERIOR adds a median 33% blocks and HOLLOW-SHELL removes a median 13%, yet the views the metric is given are near-identical in every column. The cross-sections show what changed, with sealed interior air in peach. MV-KID places both within one noise floor of real (+1.0 and +0.4); V-KID places them 287 and 156 away.

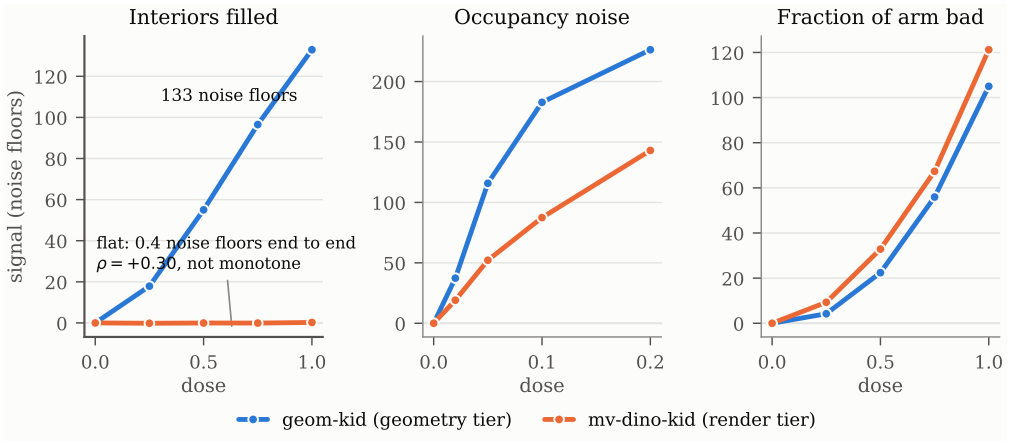


Figure 10: **Dose–response.** Each metric in units of its own noise floor, which is the only scale on which the two are commensurable. Both rise monotonically on occupancy noise and on the mixture axis; on the interior axis the render metric does not move.

The released code and data contain **no texture files**, so a reader reproducing our figures supplies them from their own installation. And the permission covers the images, not the assets: we publish

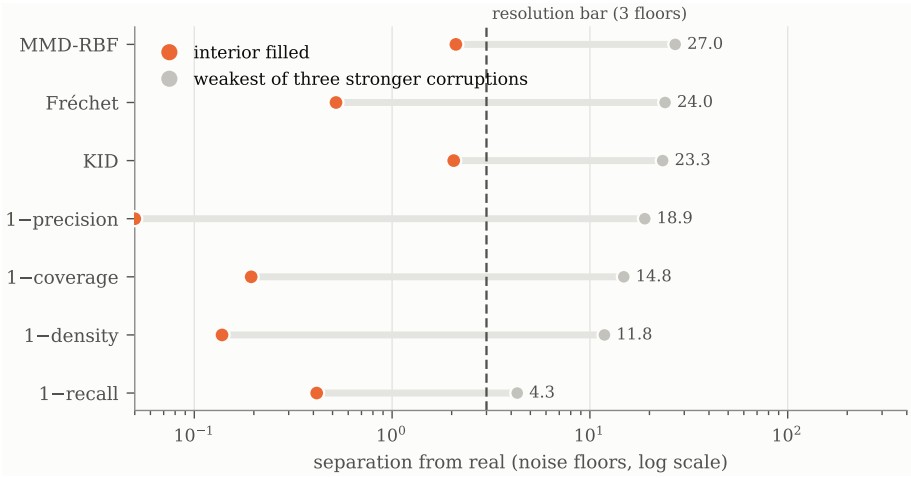


Figure 11: **Every render-space statistic shares the blindness** (LEGO, n=128). Filled marker: the interior-fill probe. Open marker: the weakest of three stronger corruptions. All seven resolve the stronger corruptions and none resolves the interior fill. One minus recall also misses the milder 20%-deletion rung at 1.4 floors, a limit of that statistic rather than of the probe.

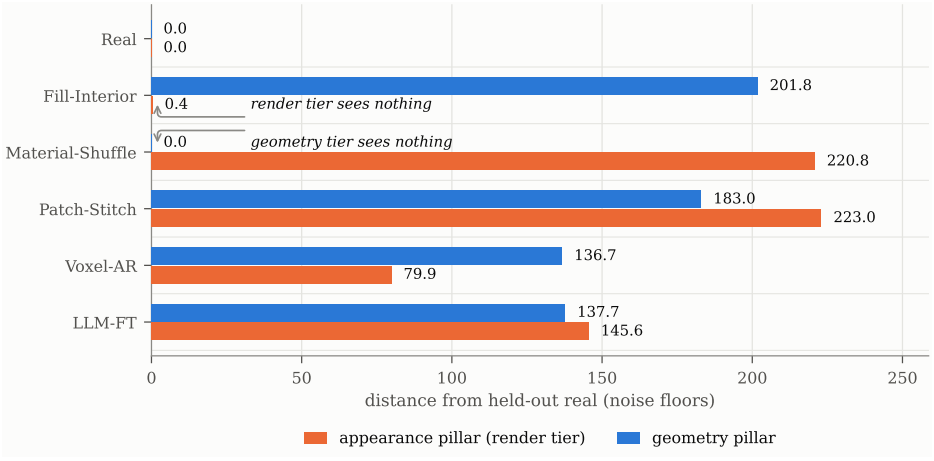


Figure 12: **The two blindnesses are mirror images.** Distance from held-out real on the two pillars. FILL-INTERIOR is invisible to the render tier and far from real on geometry; MATERIAL-SHUFFLE is the exact converse, at precisely zero on geometry.

renders, never the pack. The alternative would be flat per-block colors, which need no permission at all because they leave only third-party-authored geometry, and the figure scripts still produce them on request; we render with textures because they show what the metric is actually given. Minecraft features: © Mojang Synergies AB, included courtesy of Mojang AB. This is not an official Minecraft product and is not approved by or associated with Mojang or Microsoft.

Builds are credited by source URL in the released manifests. The original scrape did not capture uploader names, so we cannot credit individual creators, and we record that as a limitation rather than an omission. Our LEGO evaluation corpus is StableText2Brick (Pun et al., 2025), loaded from HuggingFace and not redistributed; it derives from ShapeNetCore, whose terms restrict use to non-commercial research. **[TODO: Insert the institutional determination for the corpus release and the human-subjects review before submission. Both are open questions for the university's research-compliance office, not settled by the authors.]**

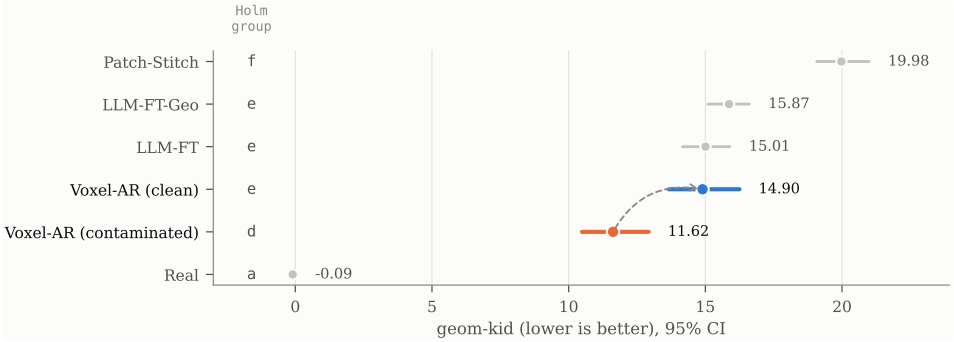


Figure 13: **The contamination case study.** V-KID with bootstrap intervals. Cleaning the split moves VOXEL-AR out of group (d) and into a tie with the baselines in group (e). The letters shown are from the original run; see Appendix O for the corrected compact letter display, under which the contaminated arm does not hold a group alone.

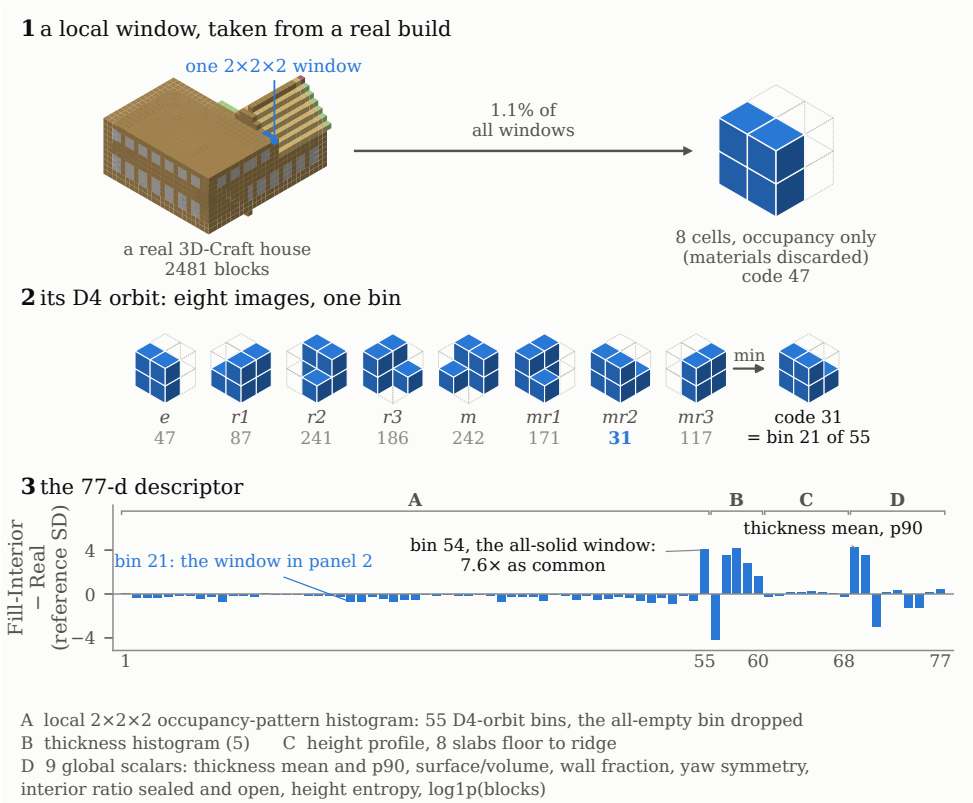


Figure 14: **The geometric descriptor.** A $2 \times 2 \times 2$ occupancy neighborhood, its orbit under the order-8 group of yaw rotations and an x-mirror collapsing to one canonical bin, and the resulting feature vector with the shift FILL-INTERIOR induces. The thickness histogram loses its one-block walls and gains thick cores, and surface-to-volume with both interior ratios falls. Filling every sealed interior moves this descriptor 202 noise floors at n=128, where the same builds move the render tier 0.4.

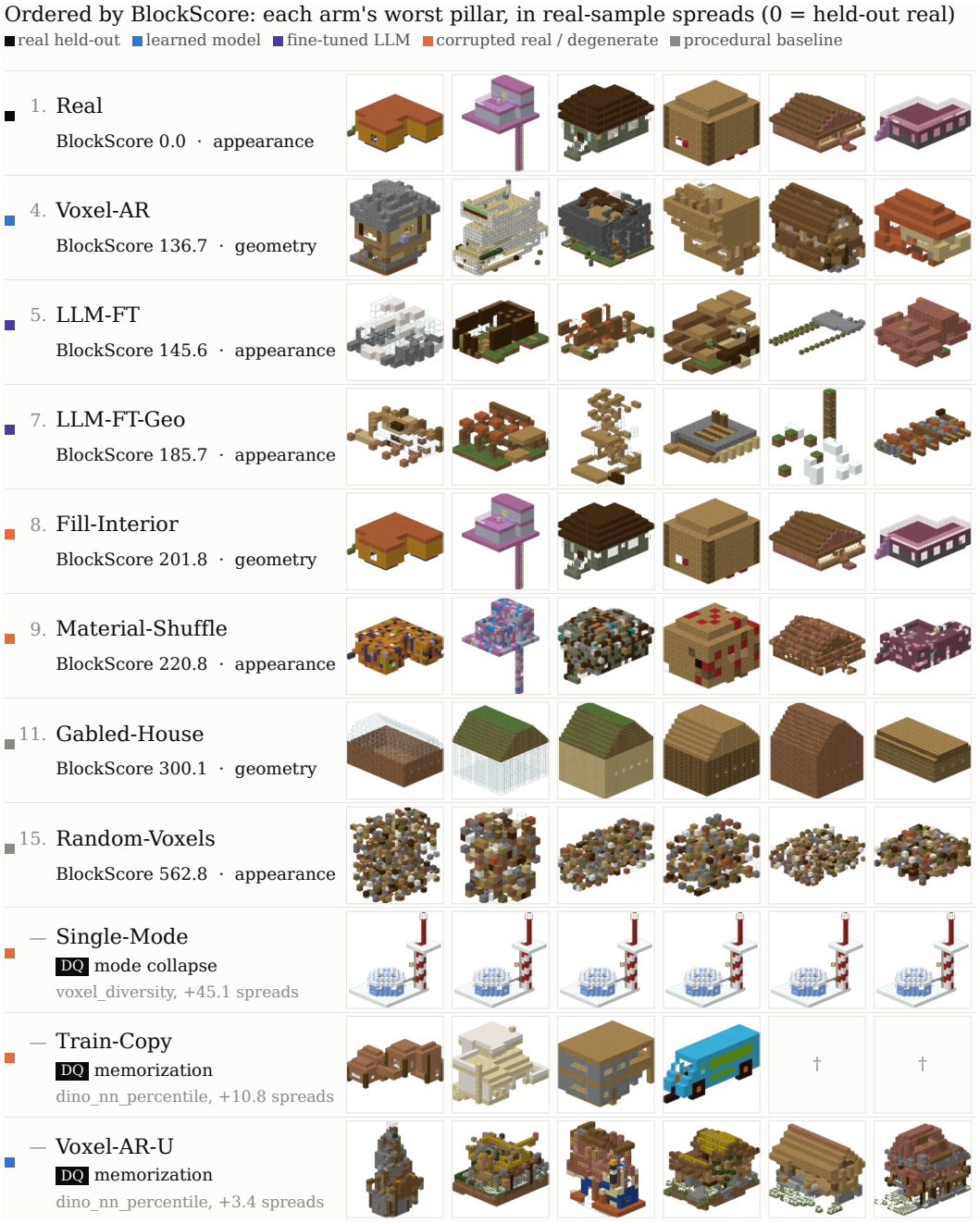


Figure 15: **Every arm, ordered by BlockScore.** Six builds per arm: the first six the run recorded, in the seeded order it recorded them, topped up from that arm's full scored sample where fewer than six remain. Builds one voxel thick are passed over, since they draw as a hairline; a fifth of what the two language-model arms emit is that shape, a truncation failure §5 describes. REAL, FILL-INTERIOR and MATERIAL-SHUFFLE are the same six buildings column for column, so a corruption reads down a column and the paper's central claim can be checked by eye: the FILL-INTERIOR row scores +201.80 and does not change. Each build is framed to its own bounding box, so cells are not to a common scale, and DQ marks an arm a disqualification gate refused a rank (Table 2). Cells marked † are withheld under GrabCraft's terms; no build shown is a memorized copy, the nearest of 1,863 training builds reaching IoU 0.51. The contaminated checkpoint of §6.5 is absent: it is unrankable, and a gallery ordered by score would imply a standing it does not have.